

Digital Transformation of Micro, Small and Medium-sized Enterprises

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ABSTRACT

This thesis explores how small and medium-sized enterprises (SMEs) approach digital transformation. The comprehensive literature review identifies gaps in the current research state. Interviews with consultants with experience in digital transformation of SMEs shed light on the problems and opportunities SMEs face from the practical perspective of experts.

Findings show that SMEs adopt a reactive, project-to-project strategy without a long-term plan. The lack of resources, both human and financial, cause problems for SMEs in all phases of digital projects. In the ideation phase, it is hard to define requirements due to the lack of in-house expertise. Employee resistance - from both management and staff - further complicates implementation. In the implementation phase insufficient financial resources force SMEs to opt for cheaper solutions. In the maintenance phase employees need to be trained, usually leading to additional consulting costs. High costs of emerging technologies compels smaller enterprises to use more mainstream and readily available technologies.

Introduction

The aim of this research is to explore the current state of digital transformation of micro, small, and medium sized enterprises. In Europe, SMEs constitute 99% of all companies in the EU [4]. With the fast changing markets and business landscape, these enterprises need to optimize their businesses and processes to stay competitive. This optimization often includes replacing human work with software or digital information systems. However, SMEs often face many challenges in their digital transformation journey, primarily due to a lack of necessary capital or expertise. Are the incentives sufficient for SMEs to commit to such an undertaking?

With this question in mind, the thesis will start by investigating the different opportunities and challenges that SMEs encounter during digital transformation, explore and address the potential risks involved, as well as look for available strategies and frameworks to approach digital transformation. This will be done by conducting a thorough literature review to assess the current state of research and determine gaps that need to be further studied.

Once these aspects have been examined, interviews with expert consultants in the field of digital transformation of SMEs will be conducted to review the most recent and effective strategies for digital transformation. The interviews will cover the initial analysis of the enterprise, the development of a transformation plan, and the final implementation and monitoring stages, all while focusing on five key topics:

- Strategy,
- Digital Technologies,
- Business Impact,
- Expertise,

- Challenges

Business impact is one of the most important aspects that this paper will look at, as it is suspected that digital initiatives are often driven by possible business expansion and profit increase. Do SMEs have access to funding, both external and internal, and is this a major road block on their path to increased digital maturity?

The term “SMEs” in this paper refers to all companies with less than 300 employees. With this in mind, how does the lack of human resources affect the digital transformation? Emerging technologies require knowledge and expertise regarding cutting-edge technologies, but with this workforce constraint, it might not be available in-house. SMEs often look for external expertise and turn to consultancy firms when trying to advance their digital capabilities, but financial problems are holding them back.

In conclusion, after conducting interviews with experts in the field of digital transformation with previous experience with SMEs, this thesis will sum up all points covered and suggest improvements on those strategies and how they can be implemented by small and medium-sized enterprise.

Methodology

The thesis is based on a dual methodology: first, a systematic literature review (SLR), followed by semi-structured interviews with industry consultants. It is, however, founded on the SLR which allows the in-depth analysis of the current academic and practical landscape relating to digital transformation in SMEs. The SLR identifies the research gaps and key themes in existing literature, informing the design of interviews in such a way that insights from consultants address under-explored areas with practical perspectives.

2.1 Literature Review

2.1.1 Formulation of the research goals

A systematic literature review begins with the definition of the research question. Since the purpose of this work is to provide a general overview of the situation of research studies on the digital transformation of SMEs at present, for a more improved systematic review of literature, the following intentions and motivations have been identified:

1. synthesis of existing literature in the field of digital transformation,
2. identification of key works and authors from the mentioned field, and
3. identification of theoretical assumptions on which strategies for digital transformation are based.

2.1.2 Locating previous studies

The second stage of the systematic review of the literature was aimed at identifying the sources in which the available literature from the area of interest will be searched. For the purposes of this research, the following online databases were used to search for articles: Springer, Google Scholar, and Elsevier.

In the above-mentioned online databases, and depending on the search options available in each of them, a search was made for titles, topics, abstracts, keywords or full texts. The following terms were used for the search: *digital transformation* OR *digital transformation in SMEs* OR *digital transformation risks and opportunities* OR *digital transformation challenges for SMEs* OR *digital maturity*.

2.1.3 Study selection and evaluation

The evaluation was done based on **28** papers included in the literature review. Since the objective of the literature review is to critically review of the current status regarding digital transformation in SME research, and to set a theoretical framework for the conduction of the interviews, the following data types were extracted from the reviewed works:

1. methodological guidelines of the work, i.e. concrete methods and techniques used for analyzing digital transformation in previous works;
2. the most significant conclusions and the key contribution of the papers in order to detect the gap in previous research;

2.2 Expert Interviews

2.2.1 Approach

This thesis discusses the digital transformation of small and medium-sized enterprises (SMEs) from the perspective of digital transformation experts, consultants. The aim of the research is to analyze how consultants see digital transformation processes within SMEs, including obstacles, opportunities, technologies used, as well as factors that contribute to the success or failure of these initiatives.

Research goals

The main goals of the interview approach is to:

1. Analyze how consultants see strategy for digital transformation in SMEs
2. Investigate which technologies the consultants consider most useful for SMEs
3. Identify the main challenges faced by SMEs during the digital transformation process, according to the consultants
4. Examine how consultants see the role of internal and external expertise in digital transformation of SMEs
5. Consider the ways in which SMEs finance their digital projects according to consulting experiences.

Reasons for this choice of methodology

Semi-structured interviews with consultants enable a deeper and more objective understanding of challenges and opportunities in the process of digital transformation of SMEs. Consultants provide a comprehensive overview of various industries and sectors, based on experience working with numerous companies. Their external perspective reduces the risk of bias, which could arise when employees from SMEs describe the transformation within their own organizations. In this way, interviews with consultants ensure a more neutral and broader insight into the key aspects of digital transformation.

Semi-Structured Interviews

The research is based on semi-structured interviews with experts in the field of digital transformation. All respondents were consultants with experience in working with SMEs, especially in connection with the implementation of digital technologies and the adaptation of digital strategies for small and medium-sized companies.

The interview format was flexible to allow adaptation according to the experience and attitudes of the respondents. The interviews covered key topics related to the strategy, technologies, challenges, expertise and financing of the digital transformation of SMEs.

Number of interviews and duration

A total of four interviews were conducted with experienced consultants who worked with different SMEs in different industries.

Each interview lasted between 45 minutes and 1 hour, and all interviews were recorded with the consent of the respondents and later transcribed.

Sampling

The sampling included four consultants from different industrial sectors, who had experience working with SMEs at different levels of digital transformation. The respondents that were selected provided different perspectives based on specializations in the following areas: IT systems, ERP, Digital Strategies, and Business Process Optimization.

2.2.2 Interview Process

The first stage started with interviewing a four sector consultants rich in experience related to implementing digital technologies for SMEs. The projects taken up by the respondents started right from idea generation and included sectors pertaining to digital transformation such as financial services, manufacturing, retail, and asset management.

- **Expert A** began his career in an online service company for developing e-commerce and knowledge bases for SMEs and works for one of the leading financial institutions as an enterprise architect.

- **Expert B** is an expert in ERP systems and Business Informatics, coming from the food supply industry. Expert B is currently working for one of the “Big 4” global accounting firms providing services in the implementation and integration of ERP technologies for SMEs.
- **Expert C** started his career in the car manufacturing industry. Expert C is a partner in a consulting firm that provides digital transformation services to both SMEs and larger enterprises. Their services are focused on the optimization of IT systems and digital strategy.
- **Expert D** has more than 25 years of professional experience operating both as a programmer and as a project manager. Nowadays, he is a consultant in how SMEs can best exploit the digital transformation to optimize their internal processes.

Specific challenges that SMEs face during digital transformation include a lack of strategic planning, employee resistance, a lack of resources, and limited access to financing. The second phase of interviews focused on these, while the consultants provided recommendations that ranged from the adoption of appropriate technologies to the development of sustainable strategies for SMEs.

2.2.3 Data Collection

The interview data collection occurred through audio recordings; afterwards, interviews were transcribed for the purpose of analysis. All participants gave consent for having them recorded and confidentiality of the data has been guaranteed. Manual data analysis of transcripts was performed to identify important patterns and themes from the consultants’ responses.

Theoretical Framework

Increased globalization in recent years has led to the occurrence of disruptive changes in the environment of almost every business. Because of their lack of large assets, both financial and human, these changes particularly affect SMEs. With the new possibility of starting a digital enterprise from the get-go, new companies enter the market more easily and effectively.

Before diving deeper into the subject matter it is important to distinguish between the terms *digitization*, *digitalization* and *digital transformation*. According to literature, *digitization* refers to “the binary representation of texts, images, sounds, films and the properties of physical objects in the form of consecutive sequences of 1s and 0s” [15]. In other words, digitization is simply the conversion of analogue data into digital form [18]. The distinction between digitalization and digital transformation is not clear and the literature is not unanimous on the definition of these terms, with them sometimes being used synonymously. However, we could say that *digitalization* focuses on the improvement of specific processes, while *digital transformation* crosses the borders between processes, companies and systems, allowing for the use of multiple digital and digitalization solutions to improve the overall functioning and efficiency of the system [28]

With this in mind, this thesis focuses on digital transformation of SMEs, and is further going to explore the opportunities and challenges that come with this process. There are different views on digital transformation, from emphasis on technology to digital customer engagement and new digital business strategies [26].

Views on digital transformation differ, ranging from an emphasis on technology to engaging customers digitally and creating new digital business strategies. Digital transformation can be categorized in three elements: 1) **technological** – based on the use of new technologies; 2) **organizational** - working on change of organizational processes and/or creation of new business models; 3) **social** - regarding digital transformation as a phenomenon influencing all aspects of human life [19]. Based on further qualitative research, literature

has identified different groups of factors influencing digital transformation in SMEs. The summary is shown in table 3.1.

Factors influencing digital transformation in SMEs [26]	
Internal factors	External factors
Capabilities fit	External capabilities fit
Resource fit	External resource fit
Changes in the business model	Government regulations
	Industry related factors

Table 3.1: Factors influencing digital transformation in SMEs [26]

Capabilities fit - a company striving for successful digital transformation must develop a large number of different capabilities [19].

Resources fit in the context of internal factors refers to a dedicated liaison device and highly authorized team [26].

Changes in the business model - Reshaping the value proposition, on which the business model is likely based, and improving interactions with customers [2] are identified as two main changes that can influence digital transformation.

As far as the external factors are concerned, it is important to note that the external factors have been a lot less explored in the existing literature [26].

External capabilities fit, and resource fit - external capabilities, e.g. collaboration or customization, as well as the historical path of the company, can be important factors in accomplishing a successful digital transformation [12].

Government regulations such as policies enabling digital uptake, supporting SME education, management skill strengthening, play a crucial role in enabling SMEs to benefit from digital transformation [16].

Industry-related factors including digital infrastructure and the possibility and eagerness to adopt new technologies contribute to the adaptability of SMEs, and therefore increase probability of successful DT [8].

3.1 Digital Technologies

A key question in digital transformation is: “Which digital technologies are currently being used and are likely to be adopted by SMEs?” Common technologies include social media, digital communication tools, online platforms, and digital marketing tools [3]. This section lists and describes additional technologies relevant to the digital transformation of SMEs.

3.1.1 Advanced Manufacturing Technology

Advanced manufacturing technology (AMT) represents different modern, computer-based systems used to improve manufacturing operations, which can lead to better positioning in the market [23]. These technologies include computer-aided design (CAD), computer-aided process planning (CAPP), computer numerical control (CNC) machines, flexible manufacturing cells/systems (FMC/FMS), and computer-integrated manufacturing (CIM)[23]. By adopting these technologies, firms can achieve quicker market entry, respond better to changing customer demands, and deliver products of higher quality with greater consistency and reliability [23].

3.1.2 Artificial Intelligence

Artificial Intelligence (AI) is most often understood as the capability of a machine to execute tasks associated with intelligent beings [5]. It may also refer to a computer or computer-controlled robot's capacity to perform cognitive functions that are often associated with the human brain, like perception, reasoning, learning, development, figuring out solutions, and interaction with the outside world. [7].

3.1.3 Augmented/Virtual Reality

Augmented reality allows a user to engage in an activity in the same physical world as without it, but adds digital information to the world that the user can interact with in the same way he can interact with the physical world. More precisely, augmented reality is a medium where information is added to the real physical world in registration with the world [6]

3.1.4 Big Data

Big Data refers to a data setup where scalable systems handle the needs of applications that process large amounts of fast-moving data in various formats. This setup can also quickly gather data at high speeds. [9].

3.1.5 Blockchain

Blockchain is a digital, distributed ledger of transactions or records, in which the ledger stores the information or data in a chain of blocks and exists across multiple participants in a peer-to-peer network [29]. Decentralisation, persistency, anonymity and auditability are the key characteristics of this technology. As transactions using the blockchain technology are decentralised, costs can be significantly decreased [29].

3.1.6 Cloud Computing

Cloud computing provides easy access to a shared pool of computing resources, which include networks, servers, storage, applications, and services, that can be easily set up

and adjusted with little management from service providers [14].

3.1.7 Internet of Things (IoT)

The term Internet of Things (IoT) is generally used to refer to the network of smart, interconnected devices that are capable of generating, exchanging and consuming data with minimal human intervention [20]. This includes objects, devices, sensors, and items that are usually not considered to be computers [20].

3.1.8 Enterprise Mobility

Enterprise mobility refers to the degree to which an organization's operations and information needs, usually related to employee activities, are supported at different locations, i.e. independently of geographical location [1].

3.1.9 Robotics

Robotics is technology that encompasses the design, construction, and use of machines to perform tasks traditionally done by humans [17].

3.2 Opportunities

Digital transformation is one of the keys for further development and growth of SMEs. Current literature identifies different opportunities DT offers these types of enterprises.

3.2.1 Economic Growth

Digitalization of business processes and digital transformation as a whole has a positive impact on economic growth [8]. It offers different advantages, such as lower entry barriers, attracting talent, and enhancing productivity and efficiency. Digital infrastructure is key for long-term financial growth of SMEs [8].

3.2.2 Innovation and Growth

Current areas of intensive research and development in technology such as blockchain, artificial intelligence (AI), machine learning are already being integrated into business models of SMEs [11].

3.2.3 Operational Efficiency

One of the main advantages and opportunities is decreasing costs and increasing operational efficiency through the adoption of new technologies in SMEs with scarce or limited resources. Using modern technologies helps to cut down on waste and increases efficiency, while enabling better usage of materials, loss reduction and increase in staff productivity. [10]

3.2.4 Competitiveness

Adopting new digital strategies can not only drive down operational costs of SMEs, but also increase competitiveness [10].

3.2.5 Resilience

Literature suggests that new technologies, such as the digital supply chain, data analytics, artificial intelligence, machine learning, robotics, digital commerce and IoT solutions can increase resilience and agility of organizations and enable them to perform better than their competitors, particularly in unstable environments [11].

3.3 Risks

While digital transformation offers so much to SMEs, it also comes with a host of risks that require special attention. This set of risks can be categorized as technical, organizational, and environmental risks.

3.3.1 Technical Risks

Technical risks are primarily about IT security. While SMEs are becoming increasingly reliant on digital technologies, they are also simultaneously exposed to cyber threats. The risk is further elevated by the limited number of IT security experts present in the labor market. In addition, the rapid rate at which technology is changing can itself be a risk because it could lead to outdated technologies being used and require continuous investment in new ones [21].

Other than these, digital technologies can be fairly complex to handle. Most SMEs lack proper technical knowledge or knowhow of these technologies, and hence cannot comprehend and use them or their full potential correctly. This results in not getting optimum value from these digital technologies, which again creates a constraint on drawing full benefits from efforts towards digital transformation. [21].

3.3.2 Organizational Risks

Organizational risks are related to the internal challenges that SMEs face during digital transformation. These would include integration of digital efforts with company's vision and goals, changes in the business model, and change management within the organization [24]. Ability to apply digital transformation independently might also be of crucial importance, since it would reduce the speed and impact of the transformation process [24].

Furthermore, digital transformation often cannot be separated from significant changes in organizational culture and mindset. SMEs may face resistance of employees who have been used to doing things in certain ways that have hardly changed throughout decades.

This might make them less willing to accept new kinds of digital technologies. This can, in turn, reduce the effectiveness of the whole digital transformation process [25].

3.3.3 Environmental Risks

Environmental risks are those that arise from external factors that cannot be anticipated or influenced by the organization. These include government regulations, industry factors, and the general economic and social environment. For example, policies that foster digital initiatives, SME education, and strengthening of management skills are critical in enabling SMEs to benefit from digital transformation.

However, these policies also vary a lot among regions and countries, hence they make the competitive playground quite uneven. Conditions of the digital infrastructure and readiness for the adoption of new technologies are more industry-related factors that can also impact the success rate of digital transformation [25].

3.4 Strategies for Digital Transformation

In general, digital transformation strategies can be categorized into those that are more suitable for SMEs and those that are more effective in large corporations. It is also worth noting that while some strategies would actually work quite well for large corporations, they would not have a positive effect, or at worst, pose a risk if applied to SMEs.

Planning for digital transformation includes both defining the strategy and deciding on the resources needed to pursue this strategy and achieve firms' goals [13]. Literature defines the Digital Transformation Framework (DTF) which is based on four transformational dimensions and their dependencies (See Figure 3.1).

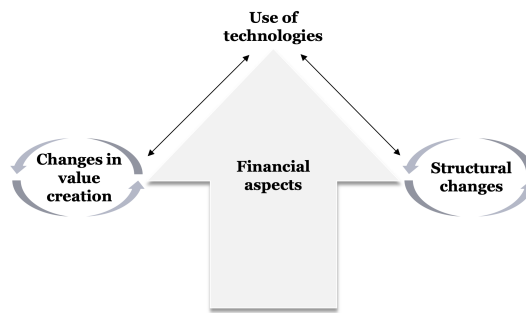


Figure 3.1: Digital Transformation Framework [13]

The four essential transformational dimensions are:

- use of technologies
- changes in value creation
- structural changes
- financial aspects

Trenkle J. later suggested establishing the term “organizational changes” instead of “structural changes”, to enhance the fit to the mindset of SME owners. It is also noted that to truly reap the rewards of digitalization, it’s important to have a clearly defined plan for integrating digital technologies [27].

For SMEs, digital transformation strategies often focus on the transformation of products, processes, and organizational aspects owing to new technologies [21]. A structured and easy-to-use approach that takes into account the challenges and opportunities of an SME, such as limited resources or high flexibility, offers a high potential for success [27].

FLOW-Agile is another approach that formalizes the visual representation of a company’s adaptive value-seeking process being implemented by Agile teams [22]. In FLOW-Agile, the goal is to make all all processes visible throughout the surfaces of operating units and let your employees collaborate in constant process redesign so that they’re always defining the best way to get the work done [22].

Other measures might be exploiting digital platforms to get resources and capabilities that are lacking internally. Digital platforms allow SMEs to have access to a wide range of resources, including software, hardware, data, and services that allow them to compete effectively against larger firms [13].

Data Analysis

4.1 The process of thematic analysis

Thematic analysis was used to code data from interviews and identify key topics related to digital transformation in SMEs. This involved the following steps:

1. **Reading and familiarization with the data:** All transcripts were reviewed in detail to identify major patterns.
2. **Identification of themes:** Key themes were identified based on respondents' responses, such as strategy, technologies, challenges, expertise and funding.
3. **Data coding:** Relevant parts of the transcripts were manually coded according to themes.
4. **Pattern analysis:** After coding, key patterns were identified that explain how the consultants see the digital transformation process of SMEs.

4.2 Key topics

The thematic analysis identified five key topics that the consultants emphasize as the most important in the digital transformation of SMEs:

4.2.1 Topic 1: Strategy for digital transformation

- SMEs often approach digital transformation without a long-term strategy. Projects are usually implemented ad-hoc, reacting to current needs and challenges, rather than planning the transformation process in advance.
- “Owners often launch digital transformation projects without a long-term strategy”.

4.2.2 Topic 2: Digital technologies

- Key technologies recommended by consultants include cloud technologies, ERP systems and automation tools. Advanced technologies like AI and big data are rarely used in SMEs due to high costs and lack of expertise.

4.2.3 Topic 3: Business Impact and financing

- SMEs rarely implement new technologies to explore new business models, and do not leverage the possibilities of emerging technologies to shift their business focus, but rather use technology to optimize and improve the current business processes and business model.
- Financing the digital transformation of SMEs relies mainly on internal resources. External sources of funding such as grants or government aid were rarely used due to the complexity of the process.

4.2.4 Topic 4: Expertise and training

- SMEs often do not have the proper internal expertise to implement digital technologies and, therefore, seek advice from external consultants. The lack of competencies and training is particularly problematic in the implementation phase of new technologies.

4.2.5 Topic 5: Challenges

- Barriers include resistance to new technologies from the workforce, organizational culture not being very welcoming to change, and a lack of resources.

4.3 Synthesis

Data synthesis indicates that usually, SMEs do not have a long-term strategic plan concerning digital transformation; the rule is often ad hoc. Consultants underline that the focus shall be drawn to core technologies like cloud services and ERP systems, though they also mention that advanced technologies are still rare due to costs and lack of expertise. The main challenges concern employee resistance and lack of resources, whereas access to external funding is limited.

Interviews/Results

This chapter presents the results from interviews with four experienced consultants who have worked with SMEs from a variety of industries. The interviews allow for a much more nuanced understanding of such key aspects of digital transformation as strategies and technologies used, challenges for SMEs, and how internal and external competencies are exploited.

To give a quick overview of the results, the findings are summarized in short in Table 5.1:

Topic	Summary
Strategy	SMEs take a "project-by-project" approach to digital transformation, with little or no long-term planning; reactivity to changes.
Digital Technologies	Focus on ERP systems, cloud services, and automation to enhance operational efficiency; cautious approach to advanced technologies due to cost and expertise limitations.
Business Impact	Digital transformation in SMEs is primarily about increasing operational efficiency within existing business models, internally funded with cautious investment as financial constraints limit access to external funding.
Expertise	Most SMEs lack the internal expertise and the resources to train in the implementation and maintenance of digital technologies, so they rely on external consultants.
Challenges	Financial constraints, resistance to change by employees and management, lack of dedicated digital strategy teams, and challenges in adapting solutions designed for larger enterprises.

Table 5.1: Overview of Topics in Digital Transformation for SMEs

5.1 Strategy

The interviewed experts were unanimous on strategy: SMEs use a “project-to-project approach” for digitization, digitalization, and the introduction of new technologies in general, with very little or no long-term strategic planning. It is about reaction rather than planning for changes. The largest catalysts of digital transformation for SMEs are external pressures such as competition or immediate technological problems. One of the interviewees explained that SMEs focus on solving specific technological issues, like automating routine tasks, without considering a broader, long-term plan.

Probably the most widespread rationale behind managing digital projects as separate initiatives is internal resource and strategic thinking deficiencies. Unlike the large enterprises that normally have a CTO or other kinds of strategic teams within their structure, SMEs are limited with regard to financial resources and human potential. This also refers to IT teams and operational managers who are usually deeply engaged in daily obligations; therefore, there hardly remains any time for planning and striving for improvement.

While that is not usually the case, SMEs that do seek development of a long-term strategy, usually do so with the help of external consultants. As mentioned in multiple interviews, formalized strategies tailored to SMEs do exist, but usually come from within the consultancy company and are specific to it. These usually include assessing current digital maturity, benchmarking against industry standards, and developing a roadmap for the future.

5.2 Digital Technologies

Across all interviews, Enterprise Resource Planning systems, cloud services, and automation tools were described as the usual starting points for SMEs. Many SMEs, especially in traditional sectors like manufacturing or food supply chain, prioritize basic tools that offer immediate operational benefits. The usual goals with these technologies are to reduce the number repetitive manual processes and to centralize data management. For example, one interviewee pointed out that some of the SMEs from past experience used outdated data management systems, with some even relying on paper records before adopting the new ERP system. Interviews also uncovered that SMEs usually choose modular and cheaper solutions, better tailored to their needs, over traditional ERP systems like SAP, better tuned for larger enterprises.

Cloud technologies are particularly popular with SMEs due to their flexibility and cost-efficiency. SMEs go for cloud services because they allow them to scale their digital infrastructure according to their needs, paying only for what they use. This is crucial for sure. Smaller businesses that cannot afford large upfront investments in IT infrastructure.

A major objective of digital transformation of SMEs is automation that allows them to cut down costs and save time. One interviewee pointed out, however, that automation

has to come with experienced business process management, as SMEs, not unlike larger companies, run a great risk of making bad processes worse by automation. Still, automating tedious and repetitive tasks, particularly in finance, operations, and receivables management, is an priority for most SMEs.

While basic technologies are receive a higher adoption rate, more advanced technologies like AI, big data analytics, and IoT are implemented cautiously. This is often due to cost and expertise barriers. One interview participant mentioned that while there is interest in AI for various tasks, these technologies are not widely used by SMEs unless a specific product is available that fits their needs. High-cost, tailored AI solutions, as well as development of a new solution in house, are often out of reach for SMEs.

5.3 Business Impact

From the interview findings, we can say that the primary focus of digital transformation in SMEs is to enhance operational efficiency rather than to redefine their business models. The exploration of new business models is rarely the driver of digital transformation. This lies on the fact that SMEs rarely have the resources to experiment with new technologies, as they see these innovation processes as hit or miss projects, resulting in either a profitability increase or a waste of money, from which they are uncertain they can recover. With this in mind, their main goal is incremental improvement of current business processes and the current business model.

SMEs generally rely on internal financial assets for funding. Funding is characterized by incremental investments, and internal financial constraints. It is not a rare case that funding is denied from top management when a project is proposed by middle management or other employees. This incremental approach thus enables the SMEs to see some benefit before moving on further, as the efficiency gains earned from the previous projects are used to finance successive projects.

While some SMEs might benefit from external sources, such as government grants or partnerships, this is usually not the case, as access to such resources is often constrained either by a lack of awareness or a shortage of expertise in how to navigate such funding programs. In some cases, these smaller enterprises reach for external expertise that can help with applications to such programs.

5.4 Expertise

The majority of SMEs usually lack the internal competencies to lead any form of digital transformation. This is often the reason why they tend to do this with the help of external consultants or by leveraging support made available by suppliers of software solutions. The reason for this can easily be explained when it comes to implementing complex systems or where there is a high degree of technical knowledge called for.

The second aspect that should not be left out is the difference between new tool capabilities and the effectiveness of their use. Many SMEs utilize only part of the features and possibilities available due to lack of understanding or training. In such cases, the full value of digital tools has yet to be realised.

The other challenge is internal knowledge transfer, since SMEs usually lack human resources that would adequately train their employees in new technologies. This often results in slow adoption rates and resistance from employees who are more accustomed to traditional methods, and the ways they used to get things done. The availability of online resources, like tutorials and webinars is often limited to mainstream, publicly available technologies.

For these reasons, SMEs typically rely on consultants and other external experts for not only the implementation of new technologies, but also for the education and training of employees. Hiring new employees who are not trained on these tools creates challenges in maintaining digital transformation projects after their completion.

As to the accessibility of external expertise, according to all interviewees this depends solely on one factor, whether an enterprise can afford it or not. The knowledge and experts are there, but the daily rates are usually too high for most SMEs.

5.5 Challenges

One of the most significant barriers to digital transformation of SMEs is the cost. Unlike larger enterprises that have the capital for the initial investment needed for more complex and innovative solutions, SMEs must carefully manage their spending. One way to solve this is the aforementioned phased implementation approach, which turns a digital transformation project into a gradual process, with SMEs implementing new tools as their budgets allow.

Furthermore, employee resistance is a recurring theme across all the interviews. Employees that have been with a company for many years are accustomed to established ways of working, and pose a hurdle to innovation and introduction of changes. This resistance is highest among older employees who fear that new technologies could make their roles in the company redundant, or their position obsolete altogether. The company culture plays a great role in overcoming this challenge, as companies who advocate change and welcome ideas coming from the bottom, have a larger chance of implementing changes successfully. It is important to distinguish between company culture and employee culture, however, as not all employees in a company that is open to innovation may embrace change. Lastly, the lack of internal departments that are dedicated to managing digital initiatives, offers lead to problems with the management of the implementation part of the process. SMEs have to invest in change and project management, as well as work on internal communication, to help carry out the process and to help employees understand the value of digital transformation and remove fears about job security.

This resistance doesn't always come from the bottom of the company; in some cases, it originates at the top. A lot of SMEs are family-owned, with the head of the company often being in their later years. Such managers are unlikely (although this is not always true) to introduce radical changes to the way things are. They are more likely to stick to incremental improvements to avoid risks and costs connected to the alteration of business processes by innovation.

Some SMEs encounter problems in the ideation and project definition phase. The reason for this can be twofold; they either don't know what is available, or they don't know what can be adapted and tailored to their enterprise. It is often the job of the consultant to engineer the requirements for the new technologies that are to be introduced. This can also be caused by scalability issues of popular solutions that are originally designed for larger enterprises. These tools and frameworks that work well for bigger corporations need to be adapted to fit the resources and needs of smaller companies. This adaptation process can be complex and requires significant effort to ensure that the digital solutions align with business processes of specific enterprises.

Conclusion

The interviews show a clearly how SMEs approach digital transformation. Their approach is reactive, with a project-to-project mentality, and they lack a comprehensive, long-term strategy. The main cause of this approach is the limitation in resources, both human and financial. While external pressures are the usual catalysts for change, most SMEs prioritize similar processes it comes to implementing new technologies; business processes which would benefit the most from automation, or the ones whose costs can be reduced immensely by innovating.

The digital tools that SMEs adopt most often include ERP systems, cloud services, and automation solutions. These technologies offer short-term benefits, which are quickly realized and tangible, such as operational efficiencies and cost reductions. However, even with these technologies, SMEs struggle to find funds and implement them because of a lack of expertise. This means they are highly dependent on external consultants and external knowledge in all phases of new digital projects - ideation and project definition, implementation, and control.

Consultants can be costly, so SMEs strive for incremental improvements and phased approaches to innovation, as this allows them to reinvest the gains from previous projects into new tools and technologies. With this cost-driven thinking, SMEs rarely implement new technologies to explore new business models, but rather focus on optimization and improvements in the procedures and processes they already have in place. Government grants and external funding seem to be scarce and highly location-dependent, so these enterprises almost solely focus on internal funding.

Additionally, employee resistance seems to be a recurring problem. This is not tied to specific parts of the company, however, as the resistance comes from both the top and the bottom. Top management can sometimes lack visionary thinking to understand the potential that emerging technologies hold, and thus limit resources and funding for innovative projects stemming from lower levels of the enterprise. Employees, on the

other hand, can feel threatened by new technologies, so the fear of job security can make them reluctant to accept changes. To solve this, SMEs should focus on cultural changes, change management, and professional project management to alleviate these problems.

In summary, SMEs recognize the potential of digital transformation, but use it mainly to improve operational efficiency, and not to innovate and expand. For those SMEs, that manage to implement digital projects successfully, the impact can easily be observed in efficiency and profitability.

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Appendix

6.1 Questionnaire for Expert Interviews

6.1.1 Strategy

Approach to Digital Transformation

- Do SMEs approach digital transformation as a series of separate projects or as a continuous, long-term process?
- Did SMEs that you've worked with before develop a long-term strategy or plan for digital transformation?
- What are the key stakeholders and sponsors from within the company that are included in the process?

Formal Approaches

- Are there specific and formalized strategies or frameworks for digital transformation adopted or created by SMEs?
- Can more complex frameworks designed for large corporations be adapted for SMEs?
- Are there usual phases an SMEs has to go through during the process? Where should they start?

Success Factors

- What factors contribute to the successful digital transformation of SMEs?

6.1.2 Digital Technologies

Adoption Patterns

- What digital technologies are most commonly adopted by SMEs? (ERP, networks, cloud data and computing, AI, VR, Big data analytics)

- Which business processes are prioritized?
- Which of the used technologies are used for which processes?
- How do Austrian SMEs differ in their adoption of digital technologies compared to SMEs in other countries?

Readiness for Advanced Technologies

- In your experience, how many small and medium enterprises are already using or interested in implementing advanced technologies such as AI, IoT, machine learning, and big data analytics? (0-25%, 25-50%, 50-75%, 75-100%)

Technology Integration

- How do SMEs integrate these digital technologies into their existing business models and workflows?

6.1.3 Business Impact

Market Advantages

- What market advantages do SMEs gain following a digital transformation?

Operational Changes

- What operational changes typically result from digital transformation in SMEs?

Impact on Business Models

- How does digital transformation affect the business model of SMEs?
- Do SMEs explore new business models enabled by emerging technologies and data utilization?
- Do SMEs use the data generated during product and service delivery to adjust their business model and increase customer retention?

Profitability and Funding

- Have you observed improvements in the profitability of SMEs that have undergone digital transformation?
- Is digital transformation usually funded through internal resources or external funding (government support, grants, tax benefits etc.)?

- What are the common issues related to funding digital transformation for SMEs?
- Would you say that funding issues arise when middle management (or an employee) proposes digital transformation to executive management, or is it the company in general that struggles with funding digital transformation?

6.1.4 Expertise

In-House Expertise

- How often do SMEs have the necessary expertise for digital transformation within their organization?
- What expertise is available during ideation, implementation and use?

Access to External Expertise

- Is external expertise accessible to SMEs?
- If SMEs require external assistance, how easy is it to obtain, considering their available resources?
- Where should they search for it, i.e. who should they rely on?

6.1.5 Challenges

Implementation and Maintenance Challenges

- What are the most common challenges SMEs face when implementing new digital technologies?
- Do these challenges include internal resistance or cultural barriers? To what extent does company culture play a role?

Transformation Obstacles

- At which stages of the digital transformation process do SMEs encounter the most obstacles?
- Do SMEs have access to resources or support to overcome these challenges effectively?

Employee Response and Training

- How do employees typically react to digital transformation initiatives?
- Is adequate training available and provided to employees to facilitate the transition?
- How does management react to digital transformation efforts? Are they involved in the process?
- How do you measure success in the end?

6.2 Interviews

6.2.1 Interview 1

Interviewer

So I believe for the data privacy purposes, I have to say here that I'm going to record this, and you should agree that it's fine. I agree. Yeah, it's only for transcribing purposes, and it's supposed to be anonymized later.

But it will be anonymized, but for demographic purposes, so I can make an analysis later, I'd like to start with, if it's fine with you, if you could tell a bit about your background and your experience with SMEs before.

Interviewee

Yeah, so I was actually, while studying at university, I was working with Austria's first commercial internet service provider. And I was responsible there for building applications for Austrian SMEs. So we were building the first online stores and internal knowledge bases.

And this is where I really learned a lot about SMEs, how they work, what their requirements are. By the way, it was amazing, we really built e-commerce stores from scratch, because there was nothing there. Totally crazy, looking at it from today.

And then I was subsequently working at a small bank, a small private bank. So we were about 10 employees, so I would say this is small. And after that, also for a small asset management company, where we were 11 or 12, something like that.

And now I'm working at a subsidiary of a large asset management firm in Austria. We are about 240 employees, which I think would also fit here.

Interviewer

Just that subsidiary?

Interviewee

Yeah, just the subsidiary. That's also an SME. Yeah, it's also an SME.

So actually, I've been working in SMEs my entire career, I just realized.

Interviewer

Great, that's great. Have you been responsible for digital transformation in a way, or just looking at people that were?

Interviewee

Responsible, not really. So in the IT service provider, I was working with the persons responsible and consulting them. Yes, you might say that in the private bank and in the small asset management company, I was responsible for IT.

So yes, it would have been actually digital transformation as well. Just to give you a time perspective. So the small bank was from 1998 to 2007.

And then the small asset manager was from 2007 to 2012. So this is a time where in the small bank, internet was still coming up. And this small bank was very closely aligned to a prominent bank at that time.

So we were using their online banking tools and everything. So in terms of digital transformation, we started on the green field, which was a good idea. So there was not a lot of transformation.

It was doing the processes right from the beginning.

Interviewer

Creation.

Interviewee

Creation. Here at the subsidiary, I'm an enterprise architect, so I have a holistic view. And one of the most important topics is digital transformation.

So how do we adopt and how do we cater to our customers with modern products, modern processes using technology. But I'm not the one deciding, for example, on budget. I'm not the product owner, for example, that would decide on which feature are we going to implement first.

Interviewer

Well, actually, an enterprise architect's view is, I guess, a lot more important for what I'm looking to answer.

Interviewee

So I would say I've always been involved in the topic. I've never been the deciding person in the end.

Interviewer

Okay. So I mentioned that I'm going to have like five different aspects that we're going to look at. I start with strategy.

And there are a lot of questions here, but I'm guessing that they're going to overlap. And as you answer, I'm going to cross most of them out. So the first thing I'd like to ask is regarding strategy.

Do SMEs approach digital transformation like from a project to project or like a continuous process where somebody is responsible for continuously improving and continuously implementing new technologies?

Interviewee

At my current enterprise, we do it from project to project. So this is definitely the topic. I have the feeling that it's actually quite hard for enterprises to have a comprehensive, long-term, stringent strategy that would really define what you typically would do with the strategy.

So where do we want to be in five years time? And now let's start projects to reach this target. I think very often it's really up to most of the time, I think, the owner who has an idea and then who tries to shape the project and the company in this direction.

I have the feeling this defining a strategy and then sticking to it is increasingly difficult for enterprises. And so we do it project by project. What we do is, for example, we have a strategy in terms of technology.

So, for example, cloud first. So whenever you're building something new, we need to do it in the cloud and we need to have very good reasons why we cannot do it in the cloud. Because we are, I don't know, we are buying some tool that's not cloud ready and needs to run on premise.

So this is acceptable then, but then we have reasons for why. So I have the feeling it's more IT strategy, some IT goals, at least in my company, that's driving the project and not so much a corporate strategy or a clear digital transformation.

Interviewer

So even at your current organization that's like...

Interviewee

They are different. Yeah. Because it's leaning towards larger and...

Yes. I think they as such has a clear strategy. I mean, I think their company software is the best example.

If they wouldn't have had a clear strategy, a corporate strategy, it would not exist today because this was a long term investment task. You don't build the software they are offering just in one project. So this group, or let's put it the way the bank within this group, they have a clear strategy.

We as an asset manager, we are kind of flowing a little bit also with what's happening on the markets. It depends. When stock markets are going good and stocks are rising, this is great for us.

We earn a lot of money and then we invest in new ideas, in new products. When interest rates go down, this is bad for us. And then strategy, yes, no, whatever.

And this is a linear relationship to our revenue. So in the year interest rates goes down, our revenue is going down. And then what we need to do is we need to cut spending.

And so it's really hard when there is such a major external factor that determines how much money you can spend to have a longer term strategy.

Interviewer

Yeah, it's really a specific example.

Interviewee

I think maybe I'm realizing this now, actually, because we are now looking at very specific products that three years, two years ago, we would not have thought about. Because we don't know how the market is going to behave. We have some ideas, we know interest rates will come down within the next couple of years.

But then again, things happen and the market is totally different.

Interviewer

OK. Now regarding, you mentioned you were consulting people that are responsible for digital transformation. I'm interested in knowing who are the key stakeholders and who are the sponsors within SMEs?

Is it the employees that come up with ideas or would be more like it comes from the top down? I mean, you mentioned it's the owners usually. But does it happen that some maybe middle management, your current position comes up with something new and then has to present it?

Interviewee

Yes, it definitely happens from my point of view. In my company, it's definitely both. So there are larger topics that are coming from the top. Some new market segments we are entering. This is clearly coming from the top where they say, guys, we need to move into this market. In our case, for example, it's digital asset management.

And this is clearly something that came from the top in alignment with the bank as well. So with our owner, they said they want to go and offer robo-advisory solutions as well. So this is how this happened.

We have lots of projects. For example, we are now working and I think this is a very nice example of digital transformation. We are currently implementing a self-service platform for our large customers.

So think about pension funds, insurance companies. This was actually driven by the sales people just out of a need. And they were really creative.

They were seeing what was driving their customers. What's driving a risk manager in a very large insurance company? What's driving a portfolio manager in a very large company?

And how could we make their life easier? And so actually for quite some time, they were collecting ideas and then coming up with a project and with the idea to implement such a platform to support the customers. So this was actually really sales employees that came up with the idea.

And they're acting kind of what in an agent organization a product owner would do. They are deciding together, so it's two or three of them, what the next feature should be that should be implemented. So I see both.

Interviewer

So both from the top.

Interviewee

So if you take an SME that's still owned by the owner or by the family, I very often see it's the owner who is driving the topics. That's my feeling.

Interviewer

And when a project does come up, are there actually formalized strategies that SMEs can use to implement them? Or is it just more without strategies? Let's just go with the flow, how it works.

Let's see. Let's start.

Interviewee

Is there a plan created? I mean, I think there are plans. The question is, do SMEs use those plans?

Do they do it? And I think I've seen both. I've seen companies that are really following a classical strategic approach.

And I think there are companies that just step by step decide and out of the blue and do it. So I think it's remarkable that there are so many methodologies you can use to follow on whatever. Even defining the strategy, then implementing, implementing this kind of projects.

And it's remarkable how I see companies that don't do this. They try to reinvent the wheel all the time. Or that really kind of in a, yeah, one step after the other.

Well, let's think about it and let's decide about it. I think the most successful companies in the end are those that have a plan and follow the plan. And that use established methodologies and tools to implement strategy and then projects.

Interviewer

That's actually what I was looking for. Great. And now your current enterprise and they obviously have some plans, some frameworks they use.

Would you say those that are like better for or designed for larger companies, can you see them being applied to smaller companies?

Interviewee

Not very good.

Interviewer

Is it a lack of people or lack of resources?

Interviewee

No, I think it's just different how you approach those type of projects or implementations. I think there's a huge dependency on the size of the company, but also on the business model of the company. This is actually something that's really challenging for us because we have to follow a project portfolio management policy.

And all the processes that are set up within the group i'm currently working at for large corporations, for large banks that are deciding on budgets that are 30, 50 million euro. And then we are here with an IT budget, change budget, maybe, I don't know, 4 million. And this just doesn't fit.

Yes, it's, for example, lack of resources. We don't have 17 different departments that contribute product management and risk and legal and whatever and market research. And I mean, in my company, they have a custom experience department.

We don't have this. And so whenever there is a process in my company that requires custom experience, we need to look at this. Or marketing, they have a specific department in the enterprise that we look at whether texts that are written for customers are understandable.

So that there are no business terms used, but general language. We don't have these people. And now they're requiring that the UX department needs to approve something.

We have no UX department. There is no UX department in our company that can approve it. And so this is just one example of where I think it really needs to fit to the size, to the culture of the company, to the business model of the company.

And it's really difficult to take models, approaches that large companies use and take them for you. You need to adjust them, adopt them, specifically to your enterprise. And I think culture is a very important topic.

Talking about digital transformation, change management is a huge topic. And this is really how is this set up in your company? Do you do change management or not at all?

Do you try to bring people along with the transformation of processes? Very often, a transformation is about replacing employees with some digital processes and tools. So how do you take care of this?

In a larger enterprise, probably this is much easier done. Because either you don't care, there's a department, you let go. Or it's easier to find other things that those people can do.

In a smaller enterprise, it's difficult. It's difficult because you know the people, you have relationships. It's not easy to let people go.

And on the other hand, it's not easy to find new work for them because there's only so much you can do in a small enterprise. So I think it really needs to be adjusted. I mean, the cornerstones will probably always be the same.

But you will need to have a different approach and an adopted approach to the size of the company.

Interviewer

So to sum up strategy, you'd say there are strategies that can be used. That usually there are plans. But that there are situations where companies just go step by step and see how it goes.

You mentioned that there is a question here about success factors. What would you say in the context of strategy that are factors that contribute to successful digital transformation? You mentioned that strategy is actually one of the success factors for successful digital transformation.

Is there something you would add?

Interviewee

I think change management. Because I mentioned it, it's really important. You need to bring people along.

Communications is always a success factor. You need to communicate the strategy. You need to communicate the goals you want to reach with this strategy.

Probably even more important than the strategy as such. I think having an understanding throughout the enterprise where the company is heading to. What it is trying to achieve.

Having this common vision. This is absolutely crucial. And then obviously making sure everyone is on board.

This is a change management topic. Communications is part of change management. I think these are the crucial topics.

You need to have the right people that fit to your culture. That fit to your organization. That understand your business.

But also that understand the new technologies you want to implement and that you want to use. I think it's virtually impossible to say, well, let's do digital transformation. I don't know.

Let's use more cloud. And you don't have any developers that understand how to build, how to do solution designing cloud environments. So you need the skills and you need obviously the resources as well.

I think today resources is a crucial topic. There's only so much you can do because you will never have enough resources. So having this clear goal.

What needs to be done? Having the resources. Making sure everyone understands.

I think these are the common stones.

Interviewer

Okay. Next would be digital technologies. And what digital technologies are most commonly adopted?

Which did you see the most?

Interviewee

I think on the forefront really of digital transformation is and needs to be the adoption of cloud technology. With everything that cloud technology really offers and brings. And there's so many topics.

One thing is use what the cloud providers are offering in services. Why is this good? Well, you can focus on really delivering business value and not do some basic commodity stuff like running infrastructure, maintaining databases or whatever.

Use the tools they are providing. Deployment tools, for example. Code repositories.

I think this is crucial. If you really do it the way it was intended, this can help a lot. Automation is crucial, I think.

So this DevOps approach, which is not a technology, but an approach how to work. I think the best technology doesn't help you if you don't use it effectively. Was it Facebook or was it Amazon?

They have, I think, 30,000 deployments a day. And this can only be done using automation, automation tools, automated deployment, automated testing. And this is crucial.

I think from the technology point of view, it's from a structural point of view. I think crucial that you don't build new monoliths because they are inflexible. Crucial today is you need to be able to adopt to changes quickly.

And so really having those small services sliced and diced in the correct way, using data streaming technology, for example. This is the important topic. In the end, I think the technology is not so crucial.

Whatever technology you choose, you should use it effectively. And you should take the time to understand how the technology you're choosing supports the goals you want to reach. But in the end, I mean, what are you referring to?

Is it the types of databases you're using or I don't know?

Interviewer

Well, usually when I just started thinking about the digital technologies that are used, I started thinking about AI, big data analytics and stuff. But it's not actually the question. The question is most of the smaller companies either don't have the resources like money or expertise to use them.

Or they don't have any need for it. I mean, for AI, obviously, most of them could find something. But my question is ERP systems, networks.

I come from Bosnia and a lot of companies don't even have a network there. And it's probably not the same in Austria. But I'm interested in when they implement those basic things at the beginning, how does it go?

Which business processes are prioritized? Which technologies are used for which segments of the company?

Interviewee

I was obviously now coming from where I am currently working. And typically in the bank, the IT department can be up to 20% of the total employees. So as I said, in Vienna, we are 240 and we have 60 employees in the IT.

It's typically for financial services, but it's not at all typically for SMEs that, for example, are in production or whatever. So they will have very few resources. And so I think that in terms of technology, whoever is responsible for the IT should really try not to do things themselves.

Whenever possible, try to find the right partners and the right tools to use. Because I think it takes a lot of time and effort and money to run all those infrastructures and tools and technologies yourself. It's really interesting.

I had a discussion the other day. I think there is no general advice. There are some companies that are outsourcing everything that's IT to externals and are just using it.

And probably more companies are doing this. But on the other hand, you have a lot of companies that are saying, well, I need to do this myself because it is important for my value creation process. I think we were discussing Amazon, which is not an SME, but they started out as an SME probably selling books.

The customer experience, the whole logistics, this is so important to them that they insource everything. I mean, I think they might now have the largest fleet or one of the largest fleets of airplanes just to move them. So this is an example.

It's important to them. It's part of their promise to the customers. If you are a prime customer, you have it in 80 percent, I think, the next day.

So they're insourcing and they do it themselves. And so generally, I think, try to do commodity things standard. Don't try to do commodity and standard things.

Running SAP or I think there's so many offerings, use them. Make sure that you pay money for resources that add value to the customer and running servers and not adding value. Unless this is part of your business.

So, again, this is a strategy comes into the game. What's contributing to my value creation process? And if it's crucial, I mean, I don't know, if I'm an IT company and I'm offering IT services, well, I probably want to run the IT infrastructure myself.

Maybe, depends. If you're a production company and you need, I don't know, an HR tool, try to get some SaaS solution for this. Because there's no value in running it.

I mean, your customers will not be more satisfied whether you run it yourself or you use just some readily available tool you're using. And if you say that, I don't know, you are delivering parts to a large number of companies and you want to make sure that this is

all automated as much as possible. And you promise them, order the part today, you'll have it there tomorrow morning.

Well, probably you would need to have some type of inventory management and delivery and even logistics systems totally integrated. Depending again on what are the services you can buy in your market. So, I think in the end, the technology is not essential.

It's really crucial. What's the strategy? What do you want to achieve?

How do you create value? And this determines how you implement it and what you are using.

Interviewer

So, you'd say that there is no general answer to the question of which business processes should be prioritized. It's basically dependent on the company.

Interviewee

I think yes. It totally depends on what's your promise to the customers. I mean, obviously, I would always try to automate and to support with digital tools, processes that are boring, that are...

Interviewer

Menial. Menial.

Interviewee

Yeah, right. I mean just also to make life for my employees easier, but then again, I mean it's a question of does it pay off, does it add to the value.

I think there's no general recommendation on which processes to automate. I think it really depends on what are you doing, what's your core business.

Interviewer

Okay, so as a consultant you wouldn't come into a company and suggest something right away without knowing specifically what you're doing and what can be. Okay, makes sense. And in your experience, now you mentioned some companies that you worked for, maybe you know some more SMEs that you didn't work for, but how many of them are actually already using the technologies that are at the forefront.

So, AI, cloud computing, IoT solutions, machine learning. That would be a question like in Austria.

Interviewee

So, the companies I worked for previously, which are rather... So, at least the private bank and which doesn't exist anymore, it was bought and the asset manager, they are not using anything of this, or they were not. But then again, with the private bank, it was dissolved, I think, in 2008.

This was not the time that you would do this. Yeah, definitely. I'm a little bit...

I have the feeling that many SMEs don't use those types of technologies. And you have mentioned the points as well, I think. There are so many preconditions that need to be established.

I mean, for AI, you need to have good quality data and a way to really access this data. I think there are some good companies that really try to do it, but then again, I think they quickly realize what's the value in the end, really. Big data analysis also requires you to have good data strategy and the setup of your data.

Which data do I collect? How do I work with it? How do I offer and provide for this data for others to use?

It really depends on how good is the company's setup. I think IoT is a topic that's being used, especially in manufacturing. Predictive maintenance, I think, is nothing to discuss anymore.

I think this is fairly established. And also, probably IoT would be the type of technology that's being used actually quite wide, I would tend to think. Most of the modern machines do have their sensors and...

In some sense, yeah, definitely. And then they can be collected. I think, as always, the huge topic is, this is very often in silos, so you have one machine from one vendor, another machine from another vendor.

Having this in a common data lake to be used, I think that's a challenge. But this is what companies are working on and what they're doing. Big data, I'm not so sure.

To be honest, I wouldn't know. And AI, I'm actually in my safe very skeptically. There are really lots of use cases that can be relevant and add value to companies.

I think this will take another couple of years, three to five years.

Interviewer

Now, I'd like to contrast this between Austria and other countries, but I'm going to come back to that when we talk about expertise. The next section would be business impact and how these digital technologies are integrated into the existing business models. Are new business models created?

What operational changes result from these technologies in SMEs?

Interviewee

I think this all comes down to what's the definition of digital transformation. Is it really following new business models or starting new business models? Amazon is such an example.

Starting on Netflix, they were sending out DVDs, now they're streaming. I think this is really true digital transformation. Very, very often, I think it's actually a digitalization of processes.

You take existing processes and you use digital tools to speed up, reduce manual work, or increase the quality, increase efficiency. Do more of the same in the same time or whatever, or be able to cater to more customers using the same resources. I think most of the projects are about digitizing processes and then improving the way things are done.

It's not so much about really entering new business models, is my feeling. Again, what I see is that once there is an alignment on what the goal should be and why some project is being done, so everyone understands what are the goals we want to reach, why are we doing it, the operationalization works well. Again, this is a topic of change management, so you need to bring business along in the project.

If you do this, it is successful when the new process is used by business. This is something where you need a lot of support from the top. It's again this alignment, this common goal, where it is clear we are going to implement something, a new way of working on a process, and then everyone is using it.

Sometimes I see, at least in my company, that we are working on a new process, doing something in a new way. We have always done it the old way, and this works, and it's set up, and I'm used to it, so I'm not going to use the new process. And then, okay, nobody cares, and then we have the same process or the same goal being achieved using two different processes.

I think this is a bad example and should not happen. This is where you need clear commitment and support from everyone, especially from the top. We are going to implement this, and by then everyone is using the new way, and you're not going to continue supporting the old way to do things.

This, I think, is important.

Interviewer

At the beginning, you mentioned that the first company you worked for, the first of the ones you mentioned, created e-commerce websites. Now, that would be an example of digital transformation for the companies that you created the web pages for. Did you then notice an increase in profitability?

Yes, absolutely.

Interviewee

So, I mean, some of the companies even exist still today. And one is actually even quite famous, and they were the first to sell their product online in Austria, and it was really quickly that they sold more using the online store than they sold in their brick-and-mortar store. Okay, that's a great example.

But again, there was the owner, and he was actually already quite old, but somehow he really wanted to have this online store. And he invested a lot of money because, as I said, everything was built from scratch. I mean, today such a product would take a couple of weeks, including payment and everything.

Back then, this was several months, because we started building the basket and then the checkout process, and he was doing the inventory management in the tool as well. Absolutely crazy, but I think it paid off. I mean, they're still existing today.

They have a good name. So, yes, most of them were successful. There are even some examples that were not successful.

Somehow they didn't follow through.

Interviewer

Of course, it happens.

Interviewee

Yeah, but most of them actually were successful.

Interviewer

Okay, I'll have to move this along. We have like 15 minutes left.

Is digital transformation in SMEs usually funded only internally, or are external funds available, like government grants, support, tax reliefs?

Interviewee

So, we are funding our projects ourselves. We tried once. I think it was through the pandemic to get some government funding, but somehow this was too cumbersome, and then we just didn't follow up on it.

I can't tell you how it is in Austria, to be honest. I don't have any information on this. So, in my company, there's no outside funding.

Interviewer

So, regarding internal funding then, what are common issues that arise in a company? Are there examples where, like we mentioned before, some middle management or an employee comes up with an idea and asks for funding from the top, and it's not granted funding?

Interviewee

So, in our case, the funding actually is quite easy. So, we need to maybe separate things. One is financing the IT department and all the developers, and the other thing is what actually activities are then done with those resources.

So, as I mentioned, we have quite a large IT department, and the IT budget basically is the financial effort to pay all those developers. This is the yearly budget we have, and obviously, yes, you have some licenses and operating costs on top, but this is marginal as compared to what you need to pay for the developers. And then the question is, what do you use those developers for?

So, which projects do you implement? How do you select them? And common practice very many years ago was that whoever was screaming the loudest or had the best connections to management, they would get their projects financed, so approved.

So, yes, you can do your project, and we will give you some part of the IT budget, so actually some of the resources, so they can implement whatever your ideas. Today, we actually have a very solid process in determining what we are using the resources for. Obviously, being in the financial industry, services industry, we do have a lot of regulatory topics, so they need to be done, and there's typically no discussion on whether there is some funding for those types of projects.

But then, obviously, and thanks God, we have many ideas of things that could be done. Now, we are here back on the strategy. You should have a long-term strategy, and thus a roadmap to implement in order to reach the strategic goals.

So, either you have this, and then you can measure against your goals how much would funding a project and implementing a project contribute to reaching the goals. And this is what we are doing. So, we really look very thoroughly into what would be the value contribution, or what would be the outcome of the implementation of a project.

And only if we are certain that a project contributes to one of the company goals, we are implementing it. This is not always very easy, to be honest. I mean, how do you know what in the end the contribution really is?

Sometimes, you really need to take assumptions, and obviously, whoever has an idea will paint the result in colorful pictures, and, ah, we are going to gain, I don't know, how many million every year more. I think this is a topic of maturity, and really also of cohesion within the company. So, we still have a little bit the feeling that everyone wants to ensure their project is done.

Maybe this is a human trait, I don't know. Everyone wants, thinks that their idea is the best, and they insist that this should be done. We probably could be better in really thinking about what is the real and true value contribution, and what's the outcome, and what is the result of a project.

So, but the times where resources or budgets were allocated just by the board saying, where you do it, they are long gone. So, this is not happening anymore. We have committees, and you need to bring in your ideas, you need to come with a business case, so you need to at least have an understanding of what the value contribution could be.

This is scrutinized, at least to some extent. So, this is how we really try to assign resources, which is the most valuable thing we have.

Interviewer

But in an even smaller company, that would still be the same.

Interviewee

I think it should be the same.

Interviewer

It should, but if you have 11 people, and the 11th one is the owner, usually the owner is still the person that says, yeah, okay, go ahead. Probably, yes. Okay.

Interviewee

I mean, I really, I think, again, it really depends on the culture, on the maturity of the company, how much this owner trusts. And yes, I think both can happen. Definitely, there are SMEs where the owner says, I want this, I don't want this, because he somehow thinks or actually maybe really does know and understand.

And I think there are even companies where people get together and really try to align on common goals and decide based on what they think is the contribution of a project or goal, what should be done.

Interviewer

Now, just one more short clarification regarding funding. We're now just talking about when there are different projects and you're deciding which ones to fund.

Interviewee

Yes.

Interviewer

But in an SME, if there's just one project.

Interviewee

Yeah. How do you get funded?

Interviewer

Yes

Interviewee

That's a good question, to be honest. I mean, we sometimes have special projects where we get and try to get additional funding. And again, it all comes down in, are we able, or is the person who thinks that this project should be done, able to communicate what the value and the contribution of a project could be.

And then in our case, you still have to go through the process. And so in, I mean, thinking about the clients we were working with in the internet service provider company, most of the time it was the owner. He said, OK.

So he had an idea. He was talking to us. We were saying, well, if you want to build this store, it's, I don't know, 40,000, 400,000K Euro.

And then he said, well, yeah, 400K Euro. Sorry. Yes, I want this implemented for me.

But then again, I know another friend of mine, actually, he's working for an SME and he's the head of finance there. And he says businesses all the time coming up with ideas, what they want to do here and there. And he keeps saying, no, we don't have the money.

Interviewer

OK.

Interviewee

So again, you cannot be generalised.

Interviewer

You can run into management that's reluctant still to implement new technologies based on funding, not just, OK. Let's move it along with expertise. Now, we mentioned that expertise in smaller companies, 10, 11, 50 people is not always available, especially for like the newer technologies, AI, cloud computing and stuff.

In which part of a digital transformation journey, like ideation, implementation and then maintenance, would you say there is the largest problem with expertise, in-house expertise?

Interviewee

I think it's implementation.

Interviewer

Implementation.

Interviewee

Yeah, definitely. Yes. OK.

Because ideation, the company needs to be able to define from a business point of view what they want to achieve and what the goals are. As I said, technology is not important most of the time. They need to be able to define what the outcome should be.

And then someone who understands and who is the expert will then decide upon which technology should be used to achieve this goal. If companies want to implement themselves, I think this is where they have the challenges. Because they wouldn't have the skills and the resources available.

Interviewer

And would you say that SMEs have external expertise available in Austria, let's say?

Interviewee

I would say yes.

Interviewer

Yeah.

Interviewee

I think yes.

Interviewer

In exchange for?

Interviewee

Money.

Interviewer

Yeah. Lots of money. Yeah, OK.

Interviewee

Well, we are back to the funding topic.

Interviewer

Yeah. But there are companies that can help SMEs in Austria. Are there points of contact that you'd suggest an SME to contact first or rely on in Austria that you know of?

Now, my question is, basically, is there government support with expertise? Because I see there are a lot of projects regarding digitalization, digital transformation.

Interviewee

To be honest, I wouldn't.

Interviewer

For a company with not a lot of money, honestly, if you have money, you can pay your way out of digital transformation.

Interviewee

Yeah, of course. Yeah. I mean, I think there is subsidiaries and external government funding that you could somehow acquire.

To be honest, I wouldn't know where to go and whom to get in touch with. OK. I mean, there is EWS, which is Austria Wirtschaftsservice.

Yeah, I heard of them. But this is just now out of the top of my mind. They probably could be helpful.

I have no experience with them.

Interviewer

So SMEs may have the same problem. Not knowing where to find people they need.

Interviewee

But then again, I think, I mean, one hour of internet search and you actually should be able to identify some institutions that would be able to support you, at least in getting some funding. I think it wouldn't be a problem at all to find other companies to help you out, which you then obviously would need to pay.

Interviewer

Yeah. And lastly, I'd like to talk about challenges like we covered a lot of them throughout this interview. Are there any that you'd like to add?

I here have a question specifically targeting culture. You mentioned that culture is a great challenge.

Interviewee

I think culture is a great challenge.

Interviewer

And internal resistance.

Interviewee

Yes. And this is exactly the topic. What's the culture in the company?

Is this company, is the culture in the company open towards innovation, trying new things, adopting new ideas, adopting new technologies, trying out new things? Or is it more a what we in German call *Bewahrmentalität*? So don't touch the running system.

It works. I think this is crucial. I've mentioned it already.

It's communications. It's really bringing people along on this journey, on trying out new things, without giving people the feeling or without having people to fear that this might have negative consequences for them. Which probably is not always easy.

I get it. And as we have already mentioned, if you're automating processes, this could mean that people might need to look for something else to do. I always like those stories where we say we want to give people a better job in terms of remove all those repetitive, manual, cumbersome, boring tasks and give them something to do that's maybe a little bit more challenging.

Then again, I think there are many people that don't even want more challenging jobs. But I think there are not too many. Bring people along.

It's a change management communications topic. I think the challenges is probably of small and medium enterprises. You really need to know and you need to understand how you create value.

Because larger companies will have the resources to adopt new technologies and to do transformations. And as a smaller company, I mean, I know a company, for example,

just coming to my mind. They were forced, by the way, by a government agency that was funding them to implement an accounting tool.

And they even insisted on they need to implement SAP. And this company didn't succeed in successfully implementing SAP and went bankrupt. Because then in the end, the cost for the implementation of SAP was so high that they couldn't pay anymore.

So I think you need to understand where is the value? What's the value? How do you generate this value?

You probably only have one chance to do it right. For larger companies, this is not the topic. If one of the projects fails, well, it fails.

And then, OK, we have lost a couple of hundred thousands or million. Let's move on to the next topic.

Interviewer

But they'll recover.

Interviewee

They will recover. A small company will probably have huge difficulties in recovering. And also finding the niche.

I think that's a general topic in today's market. How can you compete against the large companies? Again, what's your offering?

How are you different? As compared to the larger ones that maybe are in the same industry. So I think this is a huge challenge.

Really, again, I think it comes down to strategy. What's the value your customers are expecting from you in five years from now? And I think you need to think about these topics today.

This looking forward three to five years. Where do I want to be? Because this is the time it takes to implement.

You need to think about what you need to do. You need to find maybe funding takes a couple of weeks or months until you get the approval. Then you need to implement it.

And so it takes time to implement those projects. So looking ahead, I think it's really a challenge as well.

Interviewer

And we mentioned resistance from the management.

Interviewee

Yeah.

Interviewer

But in your experience in companies you worked for, did you encounter resistance with the employees adopting new technologies?

Interviewee

I actually...

Interviewer

Did they have adequate training afterwards? Was those types of resources, were they available?

Interviewee

Probably, yeah, good enough. Good enough. Could it be done better?

6.2.2 Interview 2

Interviewer

Okay, so to start I'd like to say good morning once again and to thank you for setting the time aside to participate in this interview. I'd like to let you know that I'm gonna record this for transcribing purposes, so if you have anything against it you should say now. If not, I'd like to dive into it right away and first I'd like to start with if it's fine with you if you could tell us a bit about your background because I want to create a demographic analysis later for people that I did the interviews with.

Interviewee

Okay, of course, I'm okay with you recording this video. Okay, my name is Expert B, I studied at a university in Vienna. I studied business informatics and my background is in ERP systems and innovation, so I did my internship initially in Northern Ireland in an ERP company.

It's a software called Software B and it's essentially an ERP system for food supply chains. Afterwards, I worked in innovation as an internal consultant at the University B and since then, so since the start of this year, I've been a consultant at a large consulting company.

Interviewer

And at the first company you did your internship at, what would you say it was a small and medium enterprise? How many people did they have?

Interviewee

So at the time, I was with them a bit more than a year first as an intern and then as a sales representative. When I started, it was somewhere around 70 and by the year end, it was actually more around 120, so it was quite a quick growth.

Interviewer

Okay, and they worked with other companies selling ERP systems or?

Interviewee

Yeah, so they had clients for other companies, yeah, they were B2B.

Interviewer

And they were mostly small and medium enterprises as well or large companies?

Interviewee

No, so there was quite a diversity, so that was the benefit of the software that it was able to support businesses that had only one vendor, two businesses that were dealing with thousands of vendors on a daily basis. So we had very small businesses and also very large businesses. I assume I'm not allowed to name, but yeah, two very large businesses were also customers.

Interviewer

Okay, so to start, as I mentioned before, I'm gonna do like five different sections. I'm gonna talk about strategy, digital technologies, the business impact of digital transformation, the expertise and the availability of expertise to SMEs and lastly, we're gonna talk about challenges. So to start with strategy, in your experience with SMEs, do they approach digital transformation like as a series of different projects, like separate projects or as a continuous long-term process?

And how often do SMEs have a long-term strategy or a plan specifically for digital transformation?

Interviewee

So I think this is a very interesting and specific case because what the company that I was working for was actually selling was digital transformation. So they were selling digital transformation to other companies and we had, it was definitely an ongoing project, but there were multiple sort of approaches to it. So it was mostly from the marketing side.

There were various ways that they were trying to also sell the digital transformation as a concept to other companies and then sort of sell the software with it. And that was done usually, I still have it here at home that there's like, we have like prospects for different ways that digital transformation can be done and it was mostly what it meant for most companies and also for our companies and therefore, so they tried to switch from storing any kind of data physically or also to avoid storing it in any kind of Excel sheets or any kind of other data format that cannot be traced, that cannot be tracked and verified who made what changes and that cannot provide very accurate history and it does not have a single source of truth. So this is a very important point to create documentation with one source of truth.

So for our company that meant this sort of creating systems and having a plan to develop a system that have one source of truth data that can be tracked where all the changes can be tracked and where only the highest person in the hierarchy is allowed to make some critical changes. And for that, it was an ongoing strategy for sure.

Interviewer

And that's when working with clients, but within your company, you'd probably say that regarding digitalization and digitization, it was quite developed, but there are always new technologies to introduce and then in such a company that is still an SME, would you say that the company had for its own digital transformation a plan?

Interviewee

Yeah, so we used the software that we developed to sell to the customers, we also used internally. And so we were developing it also to be relevant for our needs and to so that we ourselves also are not reliant on redundant Word and Excel documents.

Interviewer

Okay. When talking about digital transformation inside an SME, what would you say are the key stakeholders and sponsors of digital transformation inside it?

Interviewee

Of digital transformation? Yes. So for me, just to clarify, for me, when I answer these questions, it's a bit easier to answer from the perspective of the client rather than the company that I was working for because digital transformation was sort of our product.

Interviewer

Okay, yeah, that's what I'm looking for.

Interviewee

Yeah, good. Because with us being the product owners, we were also ourselves kind of forced to be sort of the leaders in digital transformation because otherwise it wouldn't make sense to sell that to the client. But when I think about the clients, there were many very important stakeholders.

So first, of course, are the clients that usually benefit from the digital transformation in the sense that they know how to, like the key thing that we're talking about when we talk about digital transformation is data and data management, right? And so having data stored in one place is very important for efficiency. Second of all, it was very important always to investors because when I'm talking about clients that were small and medium enterprises, the sort of well with traditional transformation, when they are able to store and have coherent data, it's much easier to present to investors and to sort of deal with the data and to extract some kind of information that you can put on dashboards, use Power BI and to present what you've been doing so far. Then there are also the people who work at the company. And that's a very interesting stakeholder group because when you're doing digital transformation in on any kind of company, there are a lot of people that are going to feel redundant.

There are a lot of people who have been working at the company for years and they feel like when there is an incoming digital transformation, they're not going to be necessary in a company anymore, that they're not going to be able to deal with the new tools that are being introduced in the company for the sake of digital transformation. And those might even be the biggest stakeholder group for the company that's introducing some changes because you need to sort of figure out what are you going to do with those people? Because in every company, there are like two kinds of employees, ones that are very keen to change and they would like to see things being introduced in the company to make work more efficient.

And there are others, especially older generations, who are going to have a problem with that. And how you approach that as leadership is a very important and interesting question. And the way that I've seen it being done with our clients is that you, first of all, you have to educate.

So you have to have the workshops, you have to work with people, you have to show them. And also you have to make people understand that they will not be let go, that they are wanted, that there is a purpose to them staying, that there are nothing but good thoughts towards their place in the company.

Interviewer

Great. That's a great point. And I'm going to come back to it later.

So when you approach a new client, do you have a specific or formalized strategy that is tailored to SMEs? So like the companies, the clients that were SMEs? Or is it similar when you approach a larger company and can you adopt the more complex strategies that go for a large enterprise to a smaller company?

Interviewee

Of course. So the biggest difference between a small company and a conglomerate is what are their main concerns about. So when we're talking about the small companies, say less than 150, they're usually still concerned.

So I'm talking about the food industry where I was working, right? It's mostly concerned with the basic tasks. So do I have enough space for production? What is my logistics looking like? Who are my suppliers? Am I paying my suppliers? Am I paying taxes? Am I paying the right amount of taxes? So those kind of operational tasks are the main concern.

When you're talking about the bigger conglomerate, they're talking about, there's a lot of talking about optimizations, right? So how can I optimize my production? How can I optimize my costs?

How can I optimize my suppliers and so on? So the difference there is that the big conglomerates, they're working with a lot with data. So when you're selling a digital transformation product, they're looking sort of, it's like sounds like it's the same. They both want some kind of digital transformation product that's gonna sort of make the work more efficient, but for different things. Like the SMEs, they would like it to make their production more efficient, maybe not even like to put all their, when I say what means for them more efficient, they want all of their suppliers in one place. They don't want their suppliers to be in different folders on their desktop.

For the bigger company, it will be like, who is my best supplier? Who is like giving me the biggest return? So it's like kind of different needs.

So of course, when you're talking about digital transformation, small and medium companies, small and medium enterprises, it's more sort of about the sort of the more basic functions. And they're maybe less likely to be looking into things that are like dashboards, or they're less likely to be looking at things like any kind of efficiency metrics. So they just want to keep the business running.

And sort of the optimization is kind of still down the road. It's not the primary question. Whereas for the big companies, like we're talking about optimization much more, we're talking about dashboards much more, we're talking about percentages efficiency.

Interviewer

Okay. And what I'm also interested in is when you approach an SME, you sell your products, you make a deal with them to start implementing it. Are there usually similar phases that the company needs to go through as part of the digital transformation?

And is that strategy, is it created by your company? Or is it adopted like a more general framework that normal people could find in research papers and stuff?

Interviewee

So can you repeat the question once again? So I'm sure I understand what you mean.

Interviewer

Are there specific phases that an SME has to go through after you made a deal with them to implement your software? Where do you start?

Interviewee

Right. I mean, so this is very interesting, because this is also where you kind of need to differentiate between S and M's. So small and medium enterprises, because it's very specific.

So what's very interesting is that the software that we were selling, it was quite modular. So it meant that I didn't have to sell a whole software to a company. I could also sell just, you know, like supplier management part of the software.

And not necessarily supplier management, tax management. I'm trying to find the English word, sorry, like metabyte management, like all the other modules. So I was able to sell to two companies, only a small module of things that they needed, like allergen management, maybe that's very specific for food.

And so when we're dealing with small companies, it was usually just, you first had to go through the interview phase to figure out what is it that they actually need. And sort of to then set up that one module, it's usually like one module that they need and to show like one or two people how you actually use it. And it sort of goes in that sense very quickly.

It also goes much slower in a way because the smaller medium enterprises, they're much more concerned about costs. So there is like a lot of back and forth going with how much everything is going to cost. When you have a bit of bigger enterprises, like, so if we talk about medium enterprises, then we need like, then there's usually some training, because there is going to be more people involved with the software, maybe there's one or two modules more.

And there is still this, the cost, I would say, is still quite a big factor. And sort of how is it, the budget, it's very important that it stays within the budget and that our implementation team stays within the budget. With big enterprises, it's in a way much more complicated and also much simpler because their budgets are quite flexible and usually they buy the whole thing.

So they just buy everything. What do you have in that software? Everything that you have, we're buying it, we're taking it.

And in that sense, it's also much sort of easier to work with them. With smaller enterprises, we're talking a lot about, you know, for this year, the budget is constrained. Maybe next year we can add another module.

So there is a lot of sort of also sectioning of the work as to, depending on when they're going to be able to afford what. You know, with the bigger companies, they're like, you know, here is a million. We want it all.

Interviewer

Okay. That was exactly my question. That's great.

Next, I'd like to talk about digital technologies. Now, I know you and your company sold ERP systems, but you probably noticed the digital state in which the companies, so the clients found themselves in. Do you know which other technologies the companies used?

So for example, networks, cloud and data computing, AI, VR, big data analytics, or you just did your part and sold the software you worked with?

Interviewee

So the companies that we talked to, so before they acquired our software, they usually had horrible data management systems. So low to non-existent, let's put it that way. In the most extreme cases, they didn't even have anything digitized in the computers, they were writing things on paper.

One thing above that was when they were storing things in Word documents. And maybe one thing above that is storing things in Excel sheets. So that was usually the part because the reason why that is, is that food industry is a very old industry.

It's a very slow industry. It's not really keen in changing. There are a lot of key players.

It's already dominated by certain players and things change very slowly. And also the food industry is not keen really, usually in purchasing traditional, so-called ERP software, like the more general kind of software, because it's a very, very specific industry. It's extremely regulated.

And because the industry is so extremely regulated, you will unlikely find it using things like Oracle products or SAP. So they're looking for something that's a bit more specific. And it's very likely that they will not have anything at the point when we start working with them.

And I mean, that was the beauty of the software that we were selling is just how adapted it was to the industry.

Interviewer

Okay. And you mentioned it, but I just want to reiterate, my question here that I have is which business processes are prioritized? And for SMEs, you said it would be production and supplier management, like you would put at the top.

Interviewee

Yeah. Supply. Let me think about it just real quick.

It was definitely. So again, I'm coming back to the food industry because that's the one they have the most experience in, but it was definitely supplier management.

Interviewer

Okay.

Interviewee

That was by far. Also, at the time when I was working there, there were also a lot of supply chain regulations being introduced in the EU and in the UK. So you were also responsible for things going on in your supply chain.

And if you were sourcing things that are from the other side of the world, that made things very complicated also. So supply chain, yeah, I would definitely say by far.

Interviewer

Okay. Next question would be, you probably, since they worked with ERP systems, it was probably self-explanatory, but how did SMEs integrate these digital technologies into their existing business models? Do new business models develop?

Do they change their business models or they just keep them as they were? So the business models and the workflows, how does it change?

Interviewee

For SMEs, it's usually just improvement. So it just makes things more efficient. So I wouldn't say that for SMEs there is, or actually that's not true.

The business model, at least in the case that I saw, stays the same. It just becomes more efficient, the things that they already had in mind. Of course, it opens up a lot of possibilities when you have data stored in one place and it allows you to grow.

But otherwise, I wouldn't say the business model changed.

Interviewer

Okay. But the workflow and there are operational changes, definitely?

Interviewee

Sure, of course. Because when you have simplicity, that's the key thing. It allows for a lot of simplicity.

When you implement some kind of digital transformation tool, which is in this case an ERP system, things become quicker. And because there are relatively small number of people in these companies, if you save time here, you invest that time somewhere else, which is usually in expanding the business. So, of course, the business processes became much quicker, I think, especially in the context of dealing with deliveries and suppliers, because those things are usually need to be, that's a very difficult thing.

If you think the food industry, you imagine food being delivered, every piece, those things need to be checked, those things need to be approved, those things need to be signed off on, sometimes even by multiple people. And those things take a lot of time if you do it by hand. Also, if you imagine if there are any damages to the goods, if the food is expired, if you document all of those things by hand and then also send them via email to your supplier, it's difficult to deal with all of that.

And if you imagine the product we were selling, like you have a piece of software on your phone, you take a picture, you check things that need to be checked, you click send, it's automatically sent to your supplier and you automatically get also a response from them. So, that thing speeds things up significantly. And when you have those processes being sped up, then you're investing them usually in something else, which is usually expanding the supplier network.

And then obviously your business is able to grow much quicker.

Interviewer

So, obviously this is industry dependent, but you wouldn't say that it's hard to introduce new business models in the food industry that's been established for such a long time. But you would say that somebody approached you with a question, we need an ERP system because we want to do something else, we want to change our business model. That hasn't happened?

No. Okay. So, then obviously, I don't know if you did some, after introducing your software, did you then do control and checking?

Was it successful? How did the business change? And have you observed any improvements in the profitability?

Interviewee

So, the way it worked with us is that usually our implementation team would stay with them for a while until all the employees have been educated. Sorry, I keep having German words come to my head. From nine to five, my brain switches languages.

So, yeah. So, my implementation team would usually stay with them until everyone has been sort of brought up to speed. And of course, yeah, I mean, that's the whole point we had.

I mean, I think this is public information. So, I hope so. If not, you can cut this part out of the interview.

But like, we had like insanely big customers. So, that's why the company was able to grow so quickly. It's just because also the results were very quick after implementing it.

And we were so like customers were something like Tesco in the UK was one of our biggest customers. And they sort of they implemented the software in all of their branches across the UK. It allowed them to optimize the business process quite a lot.

But they're a big company. So, I assume they're not exactly relevant. If you think about the small companies, obviously, it just allows a very quick scale up in the business operation.

Interviewer

So, it obviously gives a competitive advantage to companies.

Interviewee

Oh, absolutely. Absolutely. Yeah.

Interviewer

Makes sense. And now I have a question regarding funding, because usually with smaller companies, they don't have the resources of larger companies. Is digital transformation usually funded through internal resources or external like government support grants, then maybe tax benefits that they get?

Interviewee

This is interesting. And this I can answer you from the perspective of our own company, because we were like a digital transformation company. And we had, I mean, it still exists.

I'm just part of it. The main funding was coming from the big company behind our main client. So, it was like, we got funding from the big meat industry in the UK.

And they mostly funded the development of the software. And as sort of as a thank you, we provided that same software to all of their stores. And they were sitting in all the board meetings and stuff and deciding what's going on with the company.

That was our biggest source of income. And the second source of income was obviously the Chamber of Commerce of United Kingdom. So, they also, of course, were providing a lot of money.

So, those are the two key places that I know. I assume there could have been also some other sources of income, but the biggest ones were the big meat in the UK and the Chamber of Commerce.

Interviewer

And the clients that were SMEs, how do they fund buying your software? Did you have cases where they come up and say, this is too expensive for us, we don't have the money, or we're waiting for external funding, we're waiting for government support for something?

Interviewee

In my experience, it wasn't exactly like that. Usually the companies, because our software could be sold modularly, it very often happened that every year they would just add an additional module. So, they didn't necessarily have to have a big grant of money to finance buying our software.

It was actually quite affordable, I would say.

Interviewer

Okay. And so, how often do these SMEs, do they have the necessary expertise for the software that you get, or you provide the expertise with your software? So, you teach them everything?

Interviewee

Yeah. So, that was a very cool thing. The company hired usually for the implementation part only people from the food industry.

So, those were all people that studied some kind of food quality systems, food production, something like that. And then, within our company, and then they were sort of also taught how to work with the software when they joined their company. And then, when we worked with clients, they were able to also provide the food expertise, like as a primary thing, and then software expertise as a secondary thing.

So, yeah, we provided usually the expertise, and that was actually the selling point for the software, because it was not just a piece of software you buy, you get all the expertise with it.

Interviewer

Okay. And I don't know how integrated you were in the company of the clients, but do you know where the idea of implementing an ERP system comes from? Is it usually the top management that says, okay, we need this to optimize our processes, or is it somebody, maybe some middle management or regular employees that say, we need this to do our work better?

Interviewee

Middle management.

Interviewer

Middle management.

Interviewee

Almost, almost, yeah, middle management. There is the, sorry, middle management for the practical reasons of them needing to optimize stuff. Upper management usually is very skeptic and they're like, why am I, why would I spend any money on this? Until you pull up the tabs with the Power BI and with the nice dashboards that show them how good their business is doing or not doing. And if, once you pull up the dashboards, then the upper management is also very happy. Then they see a reason why to buy it.

But so middle management usually wants transformation for the right reasons and the upper management wants it so that they can have nice billboards.

Interviewer

Okay. And so there are cases where there's resistance from the top management.

Interviewee

Oh yeah, for sure.

Interviewer

Also regarding funding or just regarding we don't need this, like, why?

Interviewee

Both. I mean, it depends. If we're talking about small businesses, then obviously middle, upper management is also middle management and it's also much more involved and everything.

But the bigger the company gets, the more resisting, resistant the upper management gets also.

Interviewer

Okay. Excellent. And other than your expertise that you offered as a company, do SMEs hire some consultants in some cases?

External expertise, additional to help them with the changes in their processes and stuff.

Interviewee

So in our case, as mentioned, our company also kind of offered that kind of consulting with the software. So usually they wouldn't have, at least with regards to implementation of the software, they wouldn't have someone else because we can offer that's two in one combo. That's very nice for their wallets.

Interviewer

Yeah, but they're usually, like we mentioned, operational changes and maybe some people need to be reorganized, do some other work. And I'm just interested in knowing if you ever met people inside the company that are not actually from the company that were there to guide the process and help them.

Interviewee

Yeah, but they were predominantly for larger businesses. It was very unlikely to have like small SMEs have very limited budgets usually and those things cost a lot of money.

Interviewer

Okay. And the last section would be the challenges.

Interviewee

Sorry, just to interrupt you. We ourselves had collaborations with food consultants that would then recommend our software to SMEs and then they would sort of just kind of give us away to them, you know. So we had like contracts if a consultant recommends us and they take our software based on the recommendation, then they get a percentage of the sale.

Interviewer

Makes sense. And the last section would be regarding the challenges. What would you say that the most common challenges are when implementing new technologies?

Interviewee

Definitely what we talked about in the beginning, and this is especially true in the food industry, is that you don't... Implementing an ERP system is quite a big shock to the company. It's quite a big shock to the employees that have been working there for a while.

And usually in the food industry, you have people who come and stay there for 20 years. And imagine you've been doing some manual data input for 20 years and now like a 25-year-old consultant comes and says, no, no, no, now you're going to use this software.

And, you know, the people are there thinking, who are you? What are you, you know?

Interviewer

Yeah, yeah. So you would say that the culture of the company plays a big role?

Interviewee

Yeah, yeah. It's definitely, I think it definitely depends on which industry you are in. The more regulated the industry, the more difficult it is, I think, also to implement some kind of digital transformation.

Interviewer

Okay. And would you say that most challenges arise during the implementation or then afterwards when you teach the employees how to use it?

Interviewee

I think if we're talking about SMEs, they arise in the very, very beginning.

Interviewer

Okay.

Interviewee

Because a lot of them have no idea what is it exactly that they need. That's the problem.

And that's where sort of what the job of a sales person is also to kind of help them. They help them figure out what is it they need. A lot of times they come to you and they know something is not working. They know, they see that their payments are late to their suppliers.

They see that their maybe production is not as good as it could be. They see that their data gets lost. The information about suppliers gets lost. You know, they see that things are not good. But what is it exactly that they need? It's a question they don't necessarily have an answer to.

So it actually takes quite a while to align, to figure out what they need and to also align with their budget before you can even start doing anything. And the problem with working with SMEs is a digital transformation company is that they bring very, very little profit in comparison to how much effort you actually have to put in them, which also makes it, we would avoid working with smaller companies if we could, having them as clients simply because it's just, it's a lot of work with very little money in return.

Interviewer

It's not worth the struggle.

Interviewee

It's a lot of times not worth the struggle, yeah. You spend a lot of time and resources and it's not as much money as it would be if you were working for a bigger company.

Interviewer

And I have a lot of questions about employees and how it works with them. But we answered most of them. And we also mentioned how the top management usually reacts to ideas.

But are they in the end involved in the process? So usually for the small and medium enterprises, that would probably be yes, right?

Interviewee

Yeah, so for small and medium enterprises, yes, absolutely. Yeah, yeah. Then they definitely have to know what's going on.

Interviewer

Okay. And my last question would be, how do you measure success in the end?

Interviewee

I mean, that is usually, especially with SMEs, relatively simple. You sort of, you can time certain processes, how long certain things take and compare. Like how long does it take to check a delivery from your supplier and send the report about it?

If it before, before an ERP system, it took you a day and now it takes you an hour. That's already a very nice improvement. So usually you have some KPIs that you can measure, which is definitely, it depends from a company to company, what kind of module they need and what is it that they use.

But usually you measure things like the time, you measure the efficiency. So how many things you can check, because there's a lot of checks involved in the footings. How many checks can you do within an hour? How many, things like that. Depends really on the module. But usually you have some kind of KPIs that they come, that you sort of agree on in the acquisition phase. And then you sort of compare it afterwards, once you're done with setting up the software.

Interviewer

Okay, yeah. That would be all from my side. And I want to thank you again. This was great. That's why I do these interviews, some of these questions, I really didn't expect the answers that I got. That's why I do them.

Thanks again!

6.2.3 Interview 3

Interviewer

So like I mentioned, I like to record this. So if you have any troubles with that, you should speak now.

Interviewee

No, it's good.

Interviewer

Okay, great. As I said before, I'm doing a bachelor thesis on digital transformation of small and medium enterprises. And I'm having these interviews where I cover five different aspects of digital transformation. Strategy, digital technologies, business impact, challenges and the expertise that's available to SMEs. So to start, if it's fine with you, I'd like to hear about your background a bit.

Interviewee

My background, I'm 33 years into my professional life. I have started at German automobile industry for six, seven years in client server computing, as we called it in the past. I was responsible for Windows Networks, SAP systems.

And then I moved to American company located in the West Coast of the United States of America. And that company doesn't exist today, but it was taken over by Oregon in 2005 or less.

2005 was also the time when I moved to a major technology company, the second biggest US company. And I worked there for 10 years. 2015, I moved to a multinational engineering firm, that was a global player in the field of robotics, energy, transport, different divisions, some responsible for process automation in the operational space. So that was in a role of, first, operationally responsible for services. And later, I was heading transformation in the IT department, and also later in the rest of the shared services.

We called them business services at that time. And I spearheaded this transformation. My current job is at an international accounting and consulting company.

I'm a partner here, responsible for the digital transformation consulting. And my knowledge is derived from the 30 years before where I accompanied different transformation in different industries.

Interviewer

And regarding small and medium enterprises, have you worked with them before?

Interviewee

I work now with small and medium enterprise because the customer base at my current company is typically small and medium. Okay, perfect. Before, I was rather working in corporate environments.

Interviewer

Okay. So to start with the questions, not to waste any time. My first question would be, do SMEs approach digital transformation as a series of projects, or like a continuous process that goes on?

And do they have a long-term strategy usually that's dedicated to digital transformation or not?

Interviewee

Usually, so just to make sure, with SME, you mean subject matter experts compared to?

Interviewer

No, I mean small and medium enterprises.

Interviewee

Small and medium enterprises, okay.

Interviewer

So up to, let's say, 300 people.

Interviewee

Yeah, because the same acronym for different things.

Interviewer

A bit confusing, yeah.

Interviewee

So small and medium enterprises approach transformation usually in a reactive mode. There's, in most cases, that I have seen a compelling event that makes them start to think about the enterprise transformation or the small and medium business transformation. Often, I have also seen that they feel pressure from competition.

They recognize that their known competition is starting digitalization projects where they would digitize an existing process or they would transfer to another business model, which is then referred to as the transformation digitalization.

Okay, yeah, so as I said, mostly it's a compelling event and it's also pressure from the competition that we are aware of. But it's reactive in both cases. There's never some CEO that I met who says, ah, everyone talks about it and that's why I'm also doing it.

I've never seen that, yeah. I've had several conversations where people would like to explore what's possible. So they want to have a consulting on what's possible.

For example, in the real estate business, there's a lot going on with regards to automating processes. That is for sure also an exploratory approach where people say, okay, let us

understand what's out there, what's possible, and then maybe we can pick those things that we benefit from.

Interviewer

Okay, and when an SME decides to implement some new technology or something, are there any specific and formalized strategies for SMEs that they can use?

Interviewee

Yes, there are. There are specific strategies, but they are rarely formalized. So the strategies are there, yeah, but they are rather informal.

What I also see is that some business leaders, they, it really depends where they are in their professional life. When I've seen people who actually have five or six years to go until their retirement, and those people, they followed a certain strategy all their professional life. They won't deviate too much, yeah, but on the other side, people that are in the middle of building up their business, they do have a strategy, but it's not formalized.

Rarely it's formalized. It's sometimes, in one case, I've seen that the possibilities of digitalization and the need both were so high that that person has asked for a specific strategy consulting before we talk about digital and starting with digital projects. So wanted to first have a business strategy before there's a digital follow-up, yeah.

Interviewer

Okay, and what digital technologies do SMEs commonly adopt?

Interviewee

Um, usually they adopt frameworks that are sized for bigger corporations and they would strip it down to what is needed for them. Yeah. What's necessary.

Yeah, yeah, I would say so. So, yeah, most of them have their employees in mind, which is good. It would be not good to have only the business opportunities in mind.

So what I've seen is that it's not a formalized approach, but it's focused on the employees because they have a good feeling about what is possible in terms of change for the employees. They are directly connected, which is a bit the difference to corporate environments. They understand individual employees and their needs and their skills, their expectations and so on and so forth.

So they can intuitively tailor the speed of change to what's required.

Interviewer

And which business processes are prioritized?

Interviewee

Clearly financial management and controlling. That's the number one which small medium enterprises want to digitize.

It's also covering sub-processes in financial management, such as, for example, how do you say in English? Let me find a translation. Forderungsmanagement. Okay. Receivables management, accounts receivables. So when there are invoices that are not paid, that's a direct impact to the cash flow of small medium business. So that's a part of financial management, but I've seen that they are very keen to make sure that their employees don't spend too much time on repetitive tasks, such as writing, writing, how is it called? These types of business reminders about open invoices.

This is a repetitive task and it certainly has a business value, but it can be automated. Artificial intelligence can phrase up perfectly a word document that's focused on the individual situation of the one that is having the debt.

Interviewer

So that's a typical case. That's perfectly to the next question. Like specific technologies, do they use AI like you mentioned now, or is it basically just mostly ERP systems, networks?

Is there any cloud in the data computing, big data analytics, VR, have you seen that?

Interviewee

I've seen a lot in the application of cloud. Cloud is very much popular in small medium enterprises because of two reasons, in my opinion. One is that they have flexibility.

So they get only to pay what they consume, which is a cloud feature that is very popular for small medium enterprises. And secondly, if there are, let's say discontinual needs for more computing power or for more memories, they have the flexibility.

Interviewer

Yeah, it's available almost.

Interviewee

Yeah, yeah. So many of them have actually dispensed their PC under the desktop and started to subscribe to cloud services. That's a typical technology.

Then there's obviously, it depends on the business, but the more medium-sized companies, they use ERP software, Enterprise Relationship Planning software, different types of enterprise planning software. Some do open source, others do things like Microsoft Navision or SAP offerings. SAP wants to get into the SME space for almost two decades now with more than success.

So there are different types of platforms out there that make it easy for SME to leverage their core processes.

Interviewer

And these new digital technologies, do SMEs integrate them in their existing business models, or is it that they want to try out a new business model and then they try to implement new technologies?

Interviewee

It's more a gradual integration. It just happens. I haven't seen an SME that has an integration strategy on their agenda.

It just happens.

Interviewer

And does it affect the business model that they had before?

Interviewee

Often the business model is disconnected from the actual technology implementation.

Interviewer

Okay.

Interviewee

Also, we can't say it in one statement and cover all different industries. It's very depending on the industry. So if you have startups that are particular based on a digital business model, obviously it's one and the same.

You have a digital strategy and that's their model and that's it. But let's say if you have more traditional hands-on business, manufacturing or specific technologies, there is quite a disconnect between how they use automation and digitalization versus what the business is supposed to do.

Interviewer

Okay, so you wouldn't say that the change in the business model could be a driver for digital transformation? They want to implement or use a new business model and then - It's more the other way around.

Interviewee

It's the other way around, okay. They would discover things like AI and then they get the clue how they can actually change. For example, the business model impact would be I can do more volume with transactional businesses or I can gain a better ratio between senior and junior workforce and other forms of optimization.

The existing model, but that's due to their exploration of new technologies. It's really the other way around.

Interviewer

Okay. Also regarding the business impact, how do usually SMEs fund digital transformation? Are they only support external funding or it's just internal?

Interviewee

They usually fund it in a very incremental way so they would gain, I mean, there's an initial investment required, everyone does it, but then out of the increased efficiency, they try to leverage additional funds. So it's step-by-step in many iterations, if you will.

Interviewer

Yeah, so the improvements that come with new technologies, they use it for, okay.

Interviewee

Yeah, yeah. So for example, if you've got an improved customer service, you can bring down the costs that you paid before for a help desk provider, or let's assume you have engaged specialized consultancy that would do cold calls for pre-sale situation. So an agency that calls people and informs them about your product, marketing, stuff like this.

You can bring down costs, obviously, when you use social media campaigns and stuff like this.

Interviewer

And expertise, is it available to SMEs before the digital transformation, during the process, afterwards? How is it with expertise?

Interviewee

Is it available? Yes, it is.

Interviewer

Internal or external?

Interviewee

Mostly external. External. But I'm taking the question literally, which means it's available.

It's a completely different question if they would apply or use it. Availability is there. For almost each type of expertise, you find a YouTube video today, let's say.

The availability is there. The question is, how do they use it?

Interviewer

Maybe the better question would be, is it accessible, not available?

Interviewee

Yes. It is, it is. But what I can clearly see is that a lot of SMEs have their, have an existing base of assets, but rarely they use the existing functionality that is already available to them. It's just a fraction of the existing functionality that they actually use.

So a lot of the transformation program in the SME space is how the existing technology can already be utilized in order to achieve certain tasks.

Interviewer

In a better way.

Interviewee

Yeah, yeah. And that certainly also has to do with how to apply skills, how to actually be creative, think out of the box, and then use existing technology. Often they, then if there's no architecture, which is in 90% of cases, it's always no architecture.

There's no architect because they can afford an architect. So it's a homegrown set of programs and functionalities and subscriptions with more or less overlapping functionality, depending who has what preference, who orders what type of digital service. And that is then used, but only in a siloed way.

It's never seen as complete organization that needs to reduce redundancies. That is more the thinking in a corporate context. In SMEs, there's an issue, there's the next big shiny software, and then you order it.

That's it. No big software.

Interviewer

Okay, yeah. And does, let's go to the challenges. I'm running out of time already.

What would you say are the most common challenges SMEs face?

Interviewee

Number one is a limited budget. Clear. As we said, they need to refinance this out of their gains.

Number two is what we also mentioned. It is the lack of in-house expertise. They are more or less depending.

Knowledge is here, it's available, but it's hard to use it. It's hard to turn the knowledge that's externally everywhere available into something tangible they can work with. Yeah, that's the challenge.

And you can imagine there's quite resistance to change. In many of the SMEs I've seen, people that hold the strengths together, they're not necessarily on top of the hierarchy. They can be CEO minus one, minus two, and they have a lot of power because of informal relationships, because of information that they only have access to.

And so this informal power is used. And if you start to digitalize, they are challenged because then suddenly there's more transparency on the data. There's more normative processes and that all challenges their position of power.

So there's a lot of resistance to change.

Interviewer

So you would say that the culture in the company is also very important.

Interviewee

For sure. Difficulties in adopting technologies due to culture. We've always done it this way.

Our business has lived for 20 years. Why should we now change? Do you actually know what risk you're taking?

Don't do it. Don't listen to all these. Why would we need digital platforms?

Just because everyone talks about AI, no reason for adopting it.

Interviewer

And at which phase would you say that the most challenges arise? So ideation, planning, execution, phase implementation, then maintaining it afterwards and controlling?

Interviewee

I would say the main challenge is in the early stages. Yeah, it's in the early stages, clearly. The people position themselves.

They would in the early stage say, okay, I'm supporting it, I'm embracing it. Most people that try to build barriers and resist to the change, they won't officially do this. They would have a mindset like this can never work with us.

I've told them from the start, let them run into the problems and then I knew it. Then they come. I had to tell you, but I told you so.

That's the type of resistance that you face.

Interviewer

And would you say that type of resistance comes from, from which level would you say it comes from? The employees or middle management?

Interviewee

Clearly middle management. Because the transformation, you shouldn't start it if the top management isn't convinced then you better don't start it. So in one way or the other, the top management would start maybe with various unrealistic expectations, but at least they would start it.

And some of the digital changes now, especially with artificial intelligence, they come bottom up. Employees don't care if the top management has a strategy to implement AI or not. They just experiment with it and they come with it.

And as a leader, you have to deal with it if you like it or not, because your employees bring it into the company one way or the other.

Interviewer

Okay, let's say the last question. How do you measure success in the end? You, when you come in as a consultant, the whole process goes on.

And in the end, how do you measure if you succeeded?

Interviewee

Usually it's the financial numbers. So it's financial KPIs in terms of how you improve your margin, how you can make more revenue, but also it's operational efficiency, yeah. It's harder to measure success in terms of are employees more happy?

Are they more engaged? You can have employee engagement, KPIs, of course. You can measure the duration in the company.

All of it is possible, but what counts is the financial numbers. And that's it, okay. Yeah, yeah.

Interviewer

I mean- And you do see improvement in the financial numbers?

Interviewee

Of course, of course. That's usually the case. These are the hard facts that no one can deny.

All the other things you can certainly measure, qualitative, quantitative, and in the bigger corporate transformation, we do this, of course. We have lots of questionnaires, and we can assess precisely where a team is in their change curve, and we would deal with the leaders, and coach them, and all of it. But in a small and medium enterprise, what counts to them is finance and customer satisfaction.

Finance is the hard numbers. Customer satisfaction is the direct contact, qualitative assessments they would understand. Okay.

That will be it.

6.2.4 Interview 4

Interviewer

I'd like to record the meeting just so I can transcribe it later. I'd like to start, if you could tell me a little bit about your background, especially in the context of SMEs.

Interviewee

Yeah, sure. Okay, so I've been working or spending more than the last 25 years in consulting business, started my career 20 something years back as a programmer and developer, then changed to project management and when I changed to project management, something like 25 years ago, I was joining consulting business. And throughout the last 25 years, I spent actually most of my time supporting clients in the SME and enterprise level as well in technology and digitization area.

So meaning I'm a typical consultant helping clients to improve their internal processes, leveraging technology. This is what I was doing the last couple of years.

Interviewer

Great, thank you. Like I mentioned, I'm gonna cover like five different aspects. The strategy, the digital technologies that are being used today, business impact of digital transformation, expertise and challenges.

So to start with strategy, I'm interested about the approach of SMEs to digital transformation. So do they look at digital transformation as a series of separate projects or as continuous long-term process?

Interviewee

So maybe we should first use a definition what is SME for you. So when I'm talking about SME, I'm mostly talking about companies lower than 1 billion revenue. So that's the typical Austrian definition, somewhere between maybe 50, 100 million to 1 billion euro of revenue, yearly turnover.

I'm not sure if this is your definition as well, otherwise.

Interviewer

I usually talk in the terms of the number of people. So like below 300.

Interviewee

Yeah, then it's even smaller. Okay, so let me then talk only about this company size because even when you compare Germany and Austria, it's a different definition of what SME is. Okay, then.

Okay, fine. So what I see in the smaller segments of the SMEs rather on the S than the ME part of this definition is that they very often do not have a clear digitization strategy. If they start on something like what is digitization for them, they firstly look on technology, how can technology improve their daily lives in terms of, for example, single

sign-on, which is not a real part of digitization, but you know that their users, internal customers of the IT only need to sign into Outlook once and then have also their CRM system, the SAP system.

And so they don't need to re-log in again or re-authenticate. But this is where they very often start. So they very often start with coming from a technology background and having a clear technology problem.

And then only as a second step, look at the business problems. So the second step very often is coming from the finance side, where they see we have, I don't know, 10,000 invoices coming in per year and there's somebody sitting and, you know, left screen is the incoming invoice, right side is the accounting system. Somebody has to copy and paste from the PDF document something into the accounting system or the other way around.

When you send out invoices, they have to put together an email, send it out via email. So they very often come from the technology side, also using technology then to make business processes a little bit easier. But I rarely saw any strategy which said, OK, here we are today in terms of maturity level and there we want to be in five years.

Now let's start some projects, a program to transform us from here to there. It's very hard to say fire rescue driven. So where the biggest fire is, they try to solve the problem.

And throughout this dealing with technology and business problems, they find out there is more potential to increase our maturity to make things easier and then jump to the next project. So to answer your question, I rarely see a clear strategy being defined and then carried out over the course of a couple of years, but more incident management by solving a problem here, solving a problem there. Another real strategy, long term strategy followed or created by external internal people, but really solving individual problems.

Interviewer

Great. That's a perfect answer. Just, what I was looking for.

And when you say it happens like those separate projects or processes, what are the key stakeholders and sponsors from within the company that started? I'm going to talk more about the people in the organization a little bit later, but just the key stakeholders and sponsors.

Interviewee

So very often it's twofold. So it's on the one hand side, always involved is the IT department because they have the knowledge, which piece of technology they would recommend or which they would see to help in a certain situation. So it's very often IT and I'm not sure if it's 50-50 or not, but to also a big degree, it is also the business departments.

So you might be asking in terms of which sea level or which part of the organization is driving those. So I'm, for example, I'm the CDO of an SME in Austria. We have roughly 2000 people, 1 billion turnover.

So we have a CDO and it's clearly my mission, my task to run such a endeavors, to have a strategy, to kick off the projects, to supervise them, to find potential. But if you look in smaller companies, like something like two, three, four, 500 employees, maybe 500 million revenue, there's no CDO. So the ideas of how to drive digitization comes either from IT because they find a cool tool, found a cool tool, you know, not citing our chat GPT or something, but like maybe an invoice parsing scanning solution.

And they would like to convince their business departments that it would help them. Or it comes from the exactly the other way around, saying there's a CFO saying, hey, we have 10,000 of invoices coming in. I need three FTE to handle them per year.

So isn't there a technical solution to do this invoice scanning automatically? So I would say it's 50-50, but you cannot implement it without the IT, but it is only half of the time driven by IT. So I would say it's 50-50 coming from business side or coming from the IT side.

Interviewer

Okay. And when this process of digital transformation or an implementation of a new technology starts, are there specific and formalized strategies to approach it? Where do SMEs usually start? Are there specific phases that they usually have to go through?

Interviewee

There should be, but there is rarely is any. So that we're a little bit biased when interviewing myself because I was working with a consulting company and not all SMEs can afford or are willing to employ consultants. So I only had the view on SMEs, which were either big enough or open enough to hire external consultants.

So all the ones that are not working with consultants, and I would say this is 60-70%, they do whatever they do. I cannot tell you. I can only tell you what those who were involving external consultants were doing.

And there, what is the benefit of hiring a consultant? It is bringing methodology to your organization, right? You don't hire a consultant to do all of the job only.

It's also because this consultant or these consultancy companies typically have an approach of doing things. So it's hard for me to answer because what I was doing when I was helping customers to evolve their digitization strategy, I was bringing methodology with me saying, okay, this is how we typically do that things. So we were setting up a strategy by assessing the current state, defining a maturity level saying, okay, on a scale from one to whatever six, eight, 10, you know, there are different approaches out there.

We believe after having had interviews with your most important people that you are maturity level X and you are missing potential out in the following five areas. And then addressing all of these five areas with a clear roadmap of how to improve in this area. So I rarely think or how to say the other way around.

I don't think any of these clients would have had such a methodology in place. So there is nothing like a roadmap or a program or an approach of making sure to understand

where is, am I as a company today and where we want to be in five years. There's more or less, as I said, no approach.

There is simply the number of problems I have and which one is the burning one. And this one, then I address, unless you find somebody who was leaving consulting and you are hiring this person, or if you really like in my case, you are engaging with consultancy companies who bring this approach with you. And there we always followed.

We approach you, we'll never go in there like saying, okay, talking after two hours, talking to somebody saying, okay, now this is the approach to this is the problem to solve, but really first assessing the current state, projecting what would be the future state and making a clear roadmap, how to get there. And it's all based on benchmarks. So what are similar companies doing in the same size, same industry, same region, what are similar companies doing and where did they improve over time by leveraging technology?

Interviewer

So you'd say that these frameworks are rarely within the company. They don't exist there, but are brought in by the consultants that come in.

Interviewee

I never saw a company in SME, especially enterprises. Yes, but not in SMEs. I never saw an SME which had already a clear digitization approach bringing in consultants only for certain aspects.

So they were approaching me, us asking, okay, we know, we understand we have a problem with digitization. We don't know where to start. We are not sure where we are today and where we want to head to.

We need help. And then we were coming in, teaching them how to assess current state, how to project the future state and how to put together a roadmap and then supporting them going there. So yes, there was never, they have no single example where we had an SME having such a strategy in place.

And a strategy, I mean, not only a plan of projects, you know, a list of projects is not a strategy. A strategy is where are we today? Where do we want to go?

What is the way to go there and who is doing what and how do we measure it? These are the four pillars you need to know to talk about a strategy, right? Yeah.

Interviewer

Okay. And regarding digital technologies, what digital technologies are most commonly adopted by SMEs?

Interviewee

Yeah, that's a broad question because what is digital? I even today talk, discuss with colleagues with peace of mind, what is digital? What is not digital?

So there are several degrees of digitization, right? The first one is you remove the media breaks. So if you have input coming in and you need to produce output and somewhere in between there's human work needed.

Digitization aspect number one is you replace that manual work by automation. So this is the easiest way because there is tools there for decades already where you have input transformed to output. For example, again, this invoice scanning, there are since 20 years we're doing that already.

So this is the cheapest, the easiest way, maybe forget this cheapest, it's not always cheapest, but the easiest way of digitization is automation of things. Okay. This is basic.

This is what each IT department without, with or without CDO has to have as a basic skill out there. Second step is automating a bad process will result with bad automated process. So you need to have some sort of process management in place before starting to automate.

So if you have a weird process and you automate this weird process, it will be still afterwards, but it will be an automatically automated weird process. So you need to have a basic understanding of process management and in order to automate things. And the third level is leveraging technology to implement business models, which you would not be able to carry out without technology.

For example, if you are in crypto, you would not be able to run a crypto company without computation power, right? So without Nvidia graphics chips, you will not be able to do crypto mining. So this is enablement of a new business model and same goes for SMEs.

They will not be into crypto. That's clear. But if you don't have the technology to do invoice scanning, then you cannot automate it.

And the same goes with artificial intelligence solutions. For example, if you do demand forecasting or dynamic pricing, that demand for such solutions is there for 20, 30 years. You want to know how much demand will there be for a certain product in three or six months.

You would like to know it because then you can purchase more, produce more and sell it at a better price in three or six months. So the demand is there, but only now it is the time that there is a technical solution which helps you not to program for half a year or one year to achieve certain things. But you can train artificial intelligence to find the patterns for something.

So the third part is definitely understanding technology in a way or having available technology to business and new business models. And this is the how to say the newest and the most effective part at the end of the day.

Interviewer

I'm also thinking like, I come from Bosnia and I know that SMEs there, a lot of SMEs don't even use like most basic ERP systems. There's almost no digitalization at all.

So I was thinking SMEs, my thesis was supposed to be directed to Austria, but then I expanded it a bit because I didn't want it to focus so much.

ERP systems, networks, cloud data and computing, AI, VR, big data analytics. What are the usual go-tos of an SME that says I want something new, I want to change something to improve my process? Also, which business processes are prioritized and which technologies are implemented there?

Interviewee

Yeah, also depending on the industry, of course. So typically what we as consultants say, there are certain processes out there and topics out there which need to be digitized and some which are maybe less important for digitization. And the ones that are really important are those processes which are carried out several times a year, a day, a month, an hour.

So high frequency of this process with a high degree of manual interaction. So if you have a process which is carried out 10 times an hour, for example, taking calls from customers and there is a lot of people effort needed to deal with that process, that's a perfect candidate for digitization. In contrary, if you have a process like year-end reporting, which you have one person doing it for one week per year, this might not be the first point to digitize.

So there is not much gain and it's not taking place very often. So you should not start with that. So typically, and that depends on the industry, if you're in a production company, you would maybe improve on the production process.

So where to reduce the media breaks or the manual interaction without a production planning and processing. If you are in banking or small and medium and bank is maybe not important, maybe let's go for a trader or a retail company, then you would maybe look not on production, but you would look at procurements or sales planning. So typically, what is the best start is looking for a process which is carried out hundreds of times a month and which needs a high degree of human interaction.

So people work, brain, because if you look at pricing, that's maybe a reason why in Eastern Europe, not that the majority of digitization is quite low is because people are cheap there. So the hourly rate, salaries are lower than investing in companies. So rather than investing into a piece of software for 100,000 euro, you rather invest in some people hiring three, four, five people for 20,000 each.

And then you are still cheaper and more flexible than implementing a piece of software. So, but regardless where you are in the world, if you look at it for a process which is taking place hundreds of times a day or a year and which needs high degree of human interaction, this is the perfect start. Or in some industries like which are regulated, for example, you would also look at risk exposure.

So if something goes wrong, how big is the financial risk, then you would maybe also try to automate this in order to prevent human errors. But this is typically what you

would be looking for. So to be more precise, if you look at the company I'm working with at the moment, we have been automating more or less the whole finance department because we are procuring from more than 1,000 suppliers every day and we're sending invoices to our clients, hundreds and thousands every month.

So these processes had a high frequency, a lot of human interaction. We were bringing this down by automating these two processes.

Interviewer

Okay. And of these new cutting edge technologies, AI, IoT, machine learning that I mentioned, would you say that there is access? Do SMEs go for these technologies or it's just a question of resources and that they try to implement something that's maybe cheaper to implement, that the expertise is maybe more readily available?

Interviewee

I would distinguish between two levels of technology. The one is the basic technology that is provided by the big IT providers like Microsoft, Oracle, Amazon, Google and so on. So whatever they bring to the table, this is easily being adopted by IT departments because they have contracts with Microsoft, they have contracts with the database vendors.

They even have support contracts and have access to consultancy services, maybe even free of charge coming from these vendors. So if for example, you have Microsoft CoPilot, you want to see how it works, you just subscribe to Microsoft CoPilot as an admin in IT and then you can use it and play around with it in your spare time or have an intern working on that. You don't need a consulting company or project.

You can simply turn it on in your environment and try it. So this is the low level access to even very modern architectural components like AI. So CoPilot, just a click of a button and you can try and use it and find use cases for your own.

And then there are the more complex topics where there is not an out-of-the-box solution like dynamic pricing. The industry is dreaming of dynamic pricing since a couple of decades already and there is no out-of-the-box solution for that. So you need to talk to universities, you need to talk to consulting companies, to product vendors and maybe also to industry experts outside of your company and this is not being done.

So if you want to go for an AI-based dynamic pricing solution, for example, you need to hire at least two or three of these skill sets and this is not being done by SMEs because they don't have the time, not the channels, not the expertise of talking to such people. So there, they typically go for product vendors and if there is not a product vendor who is having pre-built such a solution, they will rarely access that. So I would say low-level chat GPT, CoPilot, yes, everything which is out-of-the-box access is low-level and easy.

For higher degree or high mature solutions, this is not being done very often.

Interviewer

Okay, that's the distinguishment I wanted to get to with this question.

Okay. Now, the next section would be about the business impact and I'm trying to find out how SMEs integrate these digital technologies into their business models, existing business models and do they approach digital transformation, you answered it partly in your first answer, because they are looking for a new business model. So does digital transformation enable different business models or they come from we want to change our business model, so let's implement some new technology?

Interviewee

I think rather the second option. So what I also didn't see many times was that they were saying, okay, now let's try some new technology and the possibility it brings to find a new business model and let's see what we can make out of that. This is typically the startup character within a big enterprise.

So they carve out a small team and say, now you have 1 million, play with it and bring us new business models. This is not happening in SME. What I see in SME is they have a clear problem, they search a solution for the problem and if this problem is then solved, that might be also used for different other aspects in the company.

But that they say, okay, let's now find new business models to see if this works. I never saw that. So maybe this happens without me knowing, but I doubt that there is enough budget, personal and capacity and skill set to say, let's sit together, break out, you know, we leave our daily work behind, we sit together two weeks every second week, every second we sit together for one day and brainstorm about new business models.

So I would say it's more a natural evolvement of their current business model. For example, I was working with a company who was building this, it's called Entrance Solutions. So if you go for skiing in Austria, before going up with the lift, there are such fences, gates, which you have to pass through.

We have your RFID card, put it in front and it opens or not. So I was working with one of these companies doing such entrance systems and they were saying, okay, we are the market leader for these entrance systems. What else can we open or close with our solutions in buildings or whatever?

So this was a natural involvement of, we have a core competency, what next left and right to it they can also leverage. But there was not, okay, let's go to aviation or automobiles or gastro or tourism. This was not what they were thinking or looking for.

This was more natural involvement. So to answer your question, no, I don't think that they are really speedboating new business models, startup venture capital things, but they're more trying to leverage a new technology, see, okay, this works and what else can we do with that? But not really looking for a completely new business model.

Interviewer

Okay. I also had a question here about the operational changes that typically result, but I think you mostly answered this with your answer about how you automate processes in the financial department and so on, unless you have something to add to that question.

Interviewee

No, I think that that was already summarizing. So the business impact mostly is improving the things you already do. So not inventing new things, but making things more efficient by automation.

The rest is more or less, I wouldn't say luck, but I would say needs even more attention by the top management. I'm not sure if top management in SMEs is understanding what the potential of digitization really means. Yeah.

Interviewer

And one question regarding funding, how do SMEs usually fund? Is it only through internal resources or are there some other types of financing that they use? Are there any government support grants, tax benefits?

Interviewee

So definitely there is, I would say maybe 80% is internal funds saying, okay, we have a problem. We need a piece of technology, which we then call digital, to improve the situation. And this is internally, this is more or less a business case or even not.

So sometimes even SMEs do not do a real business case. They just want to get this problem solved by technology. The other 20% do really ask for grants and funds by the state.

In Austria, there's, for example, FFG, Forschungsförderungsgesellschaft, who is really supporting companies, small and medium enterprises, by granting them funds to work on innovation. And this is also sometimes a part of the consulting business to obtain such funds from the government. So I would say maybe 20%, 15, 20% are really leveraging this already.

If you look at the last year's governmental report on innovation and FFG, you see that the funds have been more or less 80, 90% consumed by companies in Austria. So there was a limit for Austria for last year and this was more or less consumed. So it's highly being used.

Interviewer

And have you observed improvements in profitability of SMEs after?

Interviewee

Yeah, definitely. So let's distinguish also there between the basic things like having email not printed out, for example. So reducing the number of paper you waste by printing things out.

Let's do that apart for a minute because this is really, really low, lowest level of digitization. Let's talk about really process improvement automation. And there, again, it's hard to really make it general for all the SME work because I was only working for SMEs who were bringing in consultants.

So there, we were always trying to prove our results saying, okay, before we started that automation and digitization process, you had costs of, don't know, 1 million per year. After doing this project, we have brought this down to 200,000 per year. So yes, there's definitely at least where consultants are being leveraged.

There's always, how to say, a business case evaluation afterwards because you want to have a follow-up contract and prove, hey, what we did was more than successful. We just not automated. We also brought down your costs and that's the reason.

And that's the reason either you want to bring down costs or you want to generate new revenue by new business models. Second part never takes place in SME or rally. So you have to prove you brought down the costs by really analyzing that.

And, yes, I believe even if there were no consultants there, definitely there is a gain and a better cost income ratio afterwards. But I doubt that SMEs very often have the KPIs in place to measure that without the consultant. So we were very often not only delivering a digitization project, but also implementing a reporting, which is not was actually not part of the project, but we're implementing a reporting structure to prove that this was beneficial for the customer.

Interviewer

Yeah. Measuring success, basically.

Interviewee

Yeah.

Interviewer

Would you say that there are funding issues that arise when top management resists the ideas of the middle management or an employee that comes up with a new idea or requests new technologies to be implemented?

Interviewee

Yeah, we saw I saw that there are two sources of innovation ideas. It's the top down ideas where we say, OK, I heard as a top manager, I heard somewhere from a peer, other company or friend or I was on a conference and saw how successful they were. I want my company to to invest time on this as well.

And they don't have funding issues. Right. Because top management is already involved.

So this is maybe 30 percent, 20 percent of the of the projects. 70 percent of the project is not generated on top level data. The ideas are coming from the basis coming up, typically even not on middle management, but on on operational levels of people say we should automate that.

This doesn't make sense for us to do that on a manual basis. And then it's trying to be escalated upwards. And I would say, yes, once the top management is not digital native and there are rarely any digital natives on top management, not me included,

but as a CDO, no, but typically not in our company, but in SMEs, you either have the old managing directors with the white hair, you know, having been there for the last 40 years, maybe the founders of the company that it's hard to generate the funds.

If this is the second generation of the family owned business, the young and the young people taking over from their parents, then it's easier. But this is not yet there in old companies. So I would say, yes, there is definitely very often a problem to raise funds within your own company to convince your top management.

This is a good idea. There are you always also need can employing consultants to help them calculate the business case because top management is either because the idea is wonderful or I always wanted to do something like that. Don't mind the numbers.

10 percent of the times or it is so lately driven because of the KPIs. If you know, if the ROI is within one year and you have 200 percent return on invest, a 200 percent return on invest after the first year, each managing director will congratulate you and ask if you cannot be fast. So if you have the numbers at hand, you can convince, but it's hard to get the funds that you write.

And then your next question will be venture capital. No, I don't see that very often. And actually only saw it once that SME was going for external funding.

So not state driven funds. I mean, venture capital or a joint collaboration with competitors or industry followers. I only saw that once.

And this one failed because it was not the culture, not the nature of this company. OK, great.

Interviewer

We touched briefly on the expertise. I wonder if there is how often do SMEs have the necessary expertise for digital transformation in-house? You mentioned this briefly in your lecture, actually, funnily enough, how you need to work with people inside the company.

How often do SMEs have somebody inside that can help you, that understands what's happening and can help deliver this to employees?

Interviewee

So actually, there's no single person who is the digital expert. Even I am not the digital expert because you always need three elements. First one is you have to have the understanding of the of the business, right?

You have to have a real deep understanding of how the business, the functional business works. Now, I am as a consultant in my previous times or now I am as an internal employee of a wholesale company. I have a certain amount of this business understanding, but I'm even within my company.

I'm not a finance expert. I'm not a logistics expert. I'm not a supply chain.

You need somebody who is really into that business topic. Now, digitization needs also technology and the state understanding because if you don't understand what potential a certain piece of technology brings, you cannot leverage it. So if you don't understand roughly at least how AI works, you would maybe start up and say, okay, AI should for us now do, I don't know, the IT architecture.

This is a bad idea, right? So IT architecture, there can be a suggestion of AI coming out there, but it will never be driven by AI. So you need to have a good understanding of what a piece of technology can bring to you as an advantage and also disadvantage.

And these two areas on the complete opposite end of the skill level here, understand business on a very, very deep level, understand technology on a very deep level. And now the third aspect is bringing together this in a change management approach saying, okay, I understand the nerds on this side. I understand the business idiots on the other side.

Sorry for the idiots, but you know, the real experts on that side. And these are, you know, they have, there's some cliches, what IT guys and business guys look like. And the cliches are sometimes true.

You need to have somebody in between who can manage all that. So you need to have skillset to transform an organization and bring together different stakeholders. So there's no single person, and I'm not as well, who has all these three competencies.

So you need to have a team in place and form to bring this expertise to the table. Now you have IT guys. Do they have enough time to work on AI, on language models, on IoT, on all this stuff?

I'm not sure. They are good on notebooks, on servers, they're on access points and printers. There might not be enough IT guys around to talk about more than 10 minutes about AI.

On the business side, you definitely have that expertise. Otherwise, your business would not work. So you would need to hire somebody bringing expertise for IT.

And very often, there's not the expertise to transformation management. So people expertise, change management, and running projects and organizing such stuff. So very often, the skillset of project management and transformation management is completely lacking.

So the best business expertise is then charged with, okay, you are now a project manager, go for it. This will fail because the best business expertise is not the best project manager. And there, you need definitely a good project manager for such endeavors.

So to answer your question more precise, is the skillset there? In parts, yes. Business is there.

Who else should talk about your own business than the company itself? IT, in parts, is there. Infrastructure expertise is there, definitely, but maybe not AI, IoT, all the stuff around it.

And automation, process automation, is also most often not there. And what is missing most of the time is project and process management expertise, what I call the transformation management expertise. This is rarely there.

Interviewer

And if SMEs require external assistance, how easy is it for them to obtain it? Is it available? Maybe I should rephrase.

Last time I asked, is external expertise available? The answer I got was, yes, it is. But is it accessible to SMEs?

Interviewee

Accessible in terms of daily rates for getting this expertise, maybe, yes. So you can hire somebody for €1,000 a day in Austria or for €3,000 a day. The question is, can you afford it?

Do you want to afford it? Because how burning is the problem? As I said, SMEs do not set up a digitization strategy if they're not forced externally.

They do not do that. So when they want to have somebody to help them on digital aspects, they want to have a problem being solved. For example, invoice scanning.

Let's stay with this example. So then they hire somebody, either a product vendor who provides some basic sort of consultancy around that product, or they hire a consultant to select a perfect solution for them. And there, yeah, I think it is accessible on the market.

Skill set is there. It's just a matter of how much are you willing to pay. And there is enough consultants out there, either freelancers, a little bit riskier, because if they disappear within the project, they are gone.

You cannot sue somebody. Or the other side, you go to the Accenture, Capgemini, Ernst & Young of this world, which are asking for 2,000, 3,000 euros a day. This is there.

It's just a matter of money you want to afford.

Interviewer

Okay, yeah. And the last section would be challenges. What are the most common challenges SMEs face?

And in which phase of the digital transformation process do they arise most often?

Interviewee

I would say two problems, major problems. First one, we already addressed funding. How to convince the company to invest into this topic?

If there is already a burning issue, it's easier. But if you have a brilliant idea, let's go for a certain solution, because we can save some money on that. How to convince top management that it is definitely a good idea?

And it's always an investment. You cannot improve things without a prior investment, either people investment, time, or money in terms of buying a software solution or whatever is needed, or consultancies. So first problem, obtaining the internal or external funds for this idea.

And secondly, in the last quarter of the project, acceptance of people. Because what SMEs are good at are keeping a state alive, keeping things running like they are, and, you know, little fine tune here and there, but radical changes. Like we can fire two people because the invoices are automatically processed and we don't need them anymore.

This is a very crucial change to the culture. There's something new. I don't like new things.

We did it like this the last 10 years. Why should we change now? So convincing people and getting them on board for that change.

If these are the two guys which are being fired, okay, they will not be convinced, that's clear. But what I mean is if you change a process by automating it, things need to be done differently. And you also have to have trust into the new solution that this is the same or even better quality than if a person is doing that in real life.

You need to have these people working on that process in the boat to support that process. And this is very crucial and also not successful very often. So people are change resistant saying, no, we did it the way we did it the last 20 years.

Why should we change it now? I don't want this change to happen. And this is more or less maybe not the last quarter.

Maybe let's say after 50% of your process, people start sitting like this, leaning back, saying, no, this will not work. I don't want to invest my time into this because this is not a good idea. Because it's change.

It's different. It might be better, but you don't like it because it needs you to change. And this is what people very often are not willing to do.

Interviewer

So in that regard, you'd say that the culture of the company is very important.

Interviewee

I would distinguish between culture, because if you are a young startup, you are used to changes. That's clear. But it's also, you know, is the culture of the company, the culture of all employees combined?

Or is there a culture of the company and a culture of the people working in the company? I distinguish between the two because sometimes you have a company culture, which is very open, open-door policy, everyone on a first-name basis, but still there are employees below each other, amongst each other, saying not on a first-name basis, but the formal basis, or working completely different, but how their company is thinking they're working. I would say it's depending not on the culture only.

You are right, this is a certain aspect, but it's on the people. How much can the people afford to talk about change and are willing to change? If you have somebody in the company who has been there for the last 30 years, this person will definitely not be the forerunner in change.

This will be somebody who wants to protect their jobs, you know, shortly before retirement, why should they change, regardless what the company culture says.

Interviewer

And a little sub-question that wasn't originally my questionnaire. A lot of interviewees that I have done this with told me that a lot of problems arise in the actual first ideation stage where SMEs don't know what they want. Is that the case?

Interviewee

That's hard for me to answer because as I said, I was a consultant coming in when the idea was already there. So it's hard for me to understand how much time it took them to come to this idea written down, ask a consultant to join in. I would say when I look now at the last two companies, which were both SMEs in my definition, not in yours, but in my definition I was working with for the last six years, I was no more a consultant, but an employee.

In these two companies, it was either a clear burning issue we needed to address and then digital was one way of solving it, you know, bringing down the cost by automation, for example, or it was an idea coming up from operational level and it was quite clear because it was quite obvious what the problem was. It was not clear how to solve it, but it was clear what the problem was and the basic concept of how to get it sorted. So I would not agree with the previous interview saying that this was a huge problem, maybe different wording than what they meant.

What I see is when the problem was clear, it was not that easy to come to a solution, but it was always clear, okay, this is a problem and we had to somehow sort it out. So the ideation was not a problem, the concept to sort out the idea or the problem, this was very often taking a long time.

Interviewer

Okay. And I think I have like two questions left. Regarding employee training after a new technology has been implemented, in your experience, is the training handled by the consultants that come in or is it taught to a few people inside the company and then the company takes it on?

Interviewee

So let's also break it down to three areas. The first one is what is digitization and train people on open their eyes and look out for potential within their company to look for optimization or digitization potential. This was always being done by externals.

So the concept of how to train people, the content creation, the target setting, the segmentation. So this was, in my experience, always being done externally, because if you would have that competence in-house, why would you hire a consultant? Secondly, now, if there was a solution implemented, typically this has then been handed over to the business department and they were doing their training.

So it was more or less a trained trainer concept saying, okay, this is how we would train you if you would ask us. Here's the material, here's what the training should look like. We show you how you should train your people to do that.

But this was very often only a matter of cost optimization. Do we really want to have a very expensive consultant standing 10 times in front of 10 people and teaching the people the same story or do you want to do that internally? So this was done once, trained the trainer concept to the client and the client was then doing the trainings internally.

And the third one, in terms of when a real complicated piece of technology was implemented, also training the IT department of how to run it and how to maintain it. This is definitely either done by the product vendor. For example, when we were bringing in Microsoft or Google or whoever, then we hired somebody from their side as well to do the technology training for the IT departments.

So yes, I would say when it comes to technology, you at least have to upskill your SME stakeholders in maintaining that technology. But the daily operations of that piece of software typically only train once and then they handle that themselves. For example, they have also some sort of fluctuation on their side.

Even if you run that training once, half a year later there are new people coming in, you have to retrain them anyway. So that's being done typically internally only. Okay.

Interviewer

And lastly, completely opposite challenges, what would you say are the success factors of digital transformation?

Interviewee

Top-down support. So you can have a huge funnel of ideas throughout the ideation process or just because it happens. If top management is not understanding the potential and not supporting that, you would always fail either by getting funds or by getting the adoption within the company.

So for example, if you go for CRM, which some say is digitalization as well, if you implement a CRM system and you don't get support by the head of sales, it will never work. It could be an IT project failing because sales is not using CRM. And secondly, I mentioned it already before, it's the competence of supporting such a transformation.

If you have the business expertise, if you have the technology expertise, even if you have AI and data analysts, data scientists in-house, it will still fail if you don't have a department or at least one person who can carry out such a change and manage all the

necessary stakeholders. There are not that many projects like digital projects where you have business, IT, top management, even maybe banks and insurance companies brought together at one table, so external stakeholders at one table, which you need to make it a success. If you do a CRM system, you have IT and sales working together.

That's it. If you do a marketing automation, it's IT and marketing automation. But if you do a demand forecasting project, you need to have finance, you have logistics, you have to have IT, and you have your auditor in place at one table to talk together.

If this is not orchestrated very well, then it will fail. So this expertise next to funding is one of the crucial elements in running digitization projects. The best IT leader will fail in doing digitization if he has not the support by somebody being open to talk like an octopus, you know, having hands in each of the departments to all the stakeholders.

Typically, this is the role of the consultant because consultants typically, you see, talking to me, I can talk two days on such topics, you know, talk about things with people, blah, blah, here, a little bit of management there. This works for consultants quite well.

Typically, IT departments are not that open or sales experts or whatever experts are not that open.

Yeah, yeah, of course.

Interviewer

Great, that would be all, thank you!